

Upgrade In-place AVDF 20.12 to 20.15

1. List Server

- AVS01: 192.168.112.121 (HA Primary)
- AVS02: 192.168.112.120 (HA Standby)
- DBFW01: 192.168.112.178
- DBFW02: 192.168.112.179
- DBTarget : 192.168.112.232 (Database oracle)

2. Prerequisite

Download Patch number **38322746**: ISO Patch File 20.15 for AVDF

Download Patch number **35303191**: Post Upgrade Agent User Security Hardening

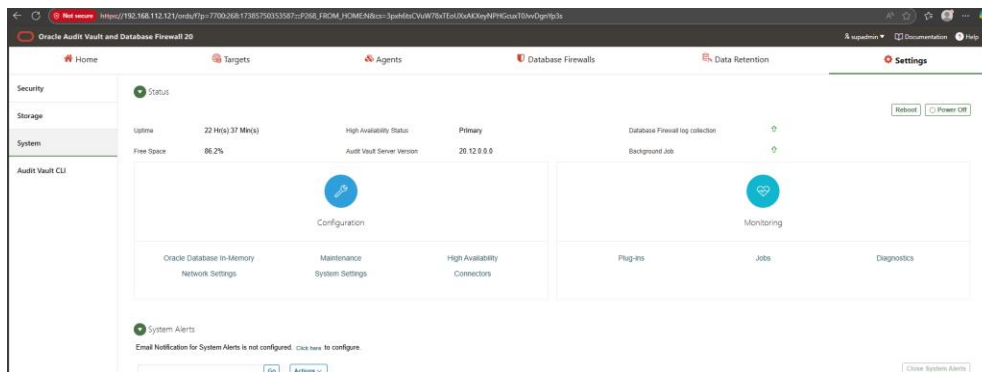
Doc ID **2458154.1**: Post Upgrade follow instructions to clear unused kernels

Download file “kernel-cleanup.rb”: follow Doc ID **2458154.1**

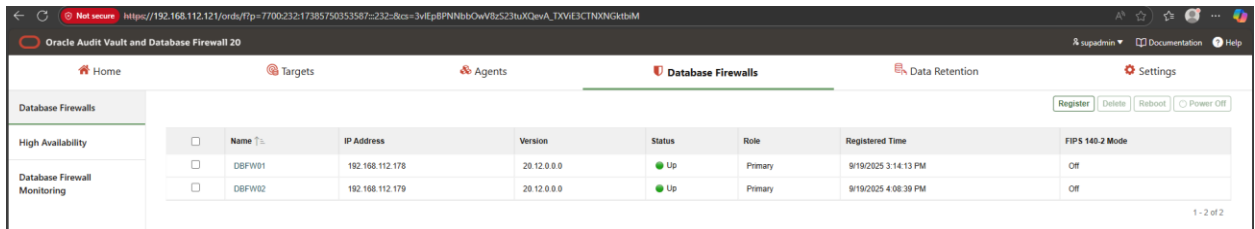
3. Precheck Before Upgrade

Before:

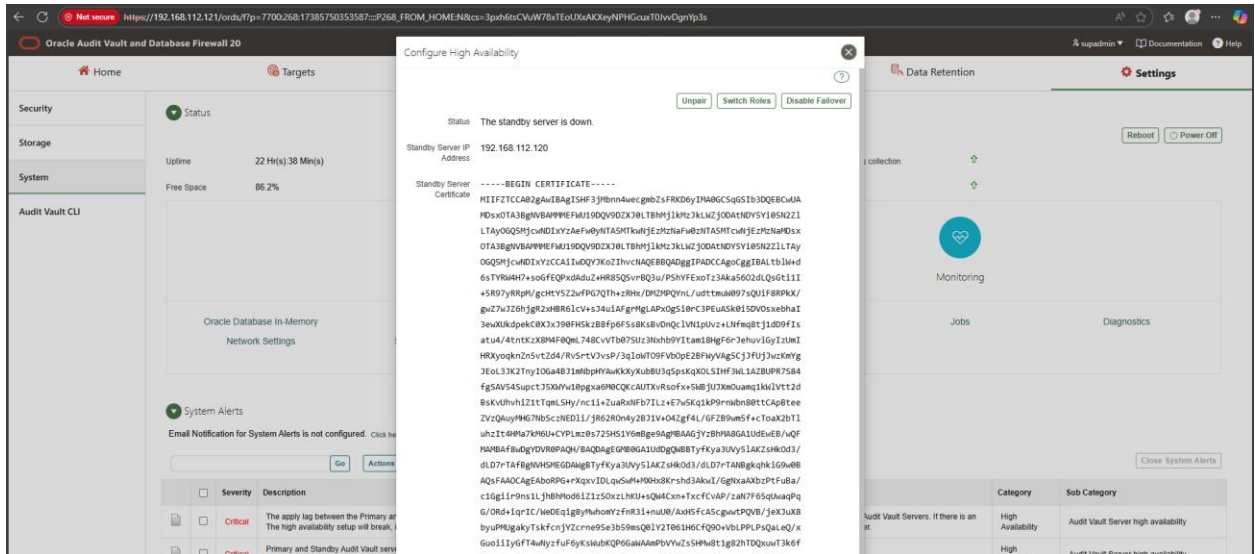
AVS Version:



DBFW Version:



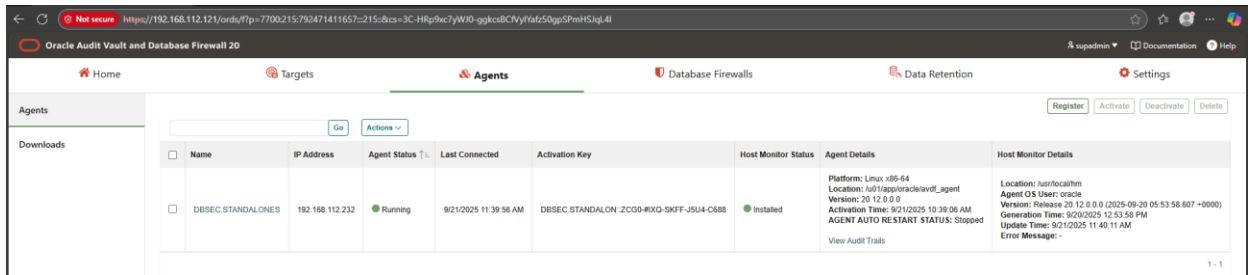
HA:



```
[oracle@avs005056a02dd1 ~]$ /usr/local/dbfw/bin/setup_ha.rb --status
HA mode: PRIMARY
HA server 1: 192.168.112.121
HA server 2: 192.168.112.120
Unique database name: DBFWDB_HA1
Current database role: PRIMARY
Data guard broker: ENABLED
Data guard observer: YES
Current log mode: ARCHIVELOG
Logging forced: YES
Flashback mode: YES
Current open mode: READ WRITE
Switchover status: NOT ALLOWED
Automatic failover: ENABLED
Failover status: TARGET OVER LAG LIMIT
Missing listener services: NONE
Archive destination: RESET
Recovery mode: MANAGED REAL TIME APPLY
FRA size: 11,811,160,064
FRA used: 2,324,692,992
FRA reclaimable: 687,865,856
FRA available: 10,174,332,928
Gap status:
Archived sequence: 31
Applied sequence: 30
Apply lag: 0:30:44
[oracle@avs005056a02dd1 ~]$
```

```
14 - rts copy
[oracle@avs005056a02204 ~]$ !12
/usr/local/dbfw/bin/setup_ha.rb --status
HA mode: STANDBY
HA server 1: 192.168.112.121
HA server 2: 192.168.112.120
Unique database name: DBFWDB_HA2
Current database role: PHYSICAL STANDBY
Data guard broker: ENABLED
Data guard observer: YES
Current log mode: ARCHIVELOG
Logging forced: YES
Flashback mode: YES
Current open mode: MOUNTED
Switchover status: NOT ALLOWED
Automatic failover: ENABLED
Failover status: TARGET UNDER LAG LIMIT
Missing listener services: NONE
Archive destination: RESET
Recovery mode: MANAGED REAL TIME APPLY
FRA size: 11,811,160,064
FRA used: 2,134,900,736
FRA reclaimable: 1,046,478,848
FRA available: 10,722,738,176
[oracle@avs005056a02204 ~]$
```

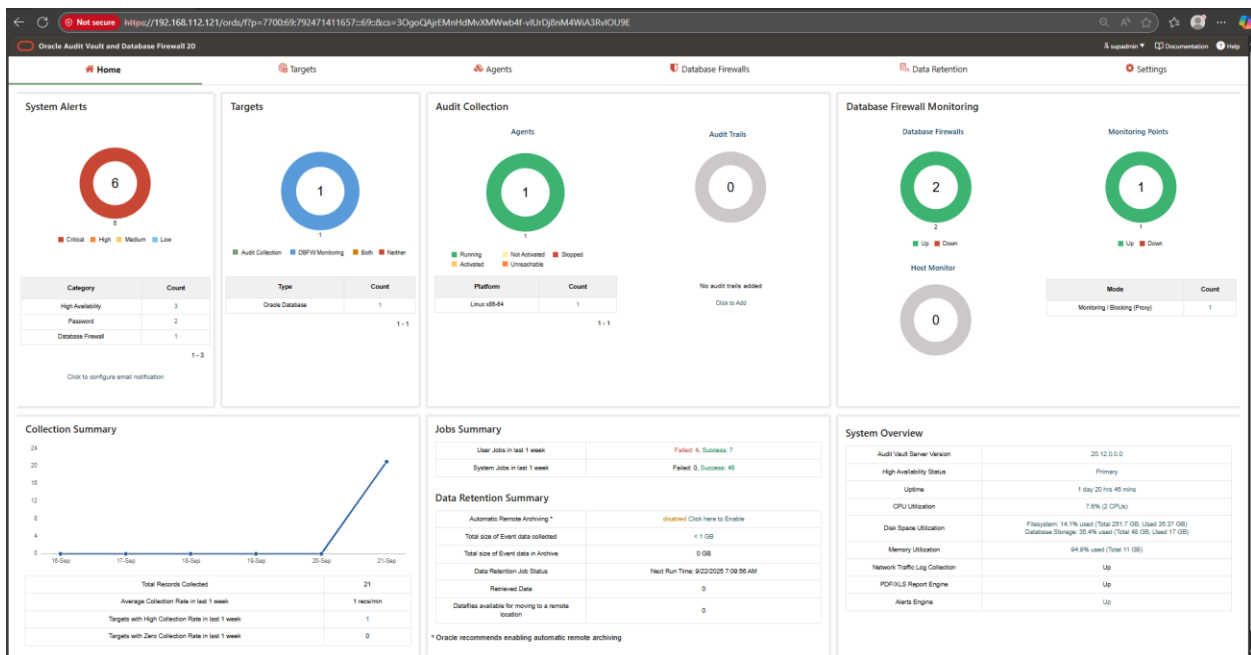
Agent Version:



```
[oracle@standalones bin]$  
[oracle@standalones bin]$ ./agentctl -v  
  
Audit Vault Agent : Release 20.12.0.0.0 (2025-09-20 05:53:58.607 +0000) - Production on Sun Sep 21 10:54:37 WIB 2025  
Audit Vault Server Connection Details:(DESCRIPTION=(ENABLE=broken)(FAILOVER=on)(RETRY_COUNT=3)(CONNECT_TIMEOUT=60)(RETRY_DELAY=3)(TRANSPORT_CONNECT_TIMEOUT=3)(ADDRESS_LIST=(LOAD_BALANCE=on)(ADDRESS=(PROTOCOL=TCP)(HOST=192.168.112.121)(PORT=1521)))(ADDRESS=(PROTOCOL=TCP)(HOST=192.168.112.120)(PORT=1521)))(CONNECT_DATA=(SERVICE_NAME=DBFWDB.UBFWDB)))  
  
[oracle@standalones bin]$  
[oracle@standalones bin]$ cat version.txt  
AV_RELEASE_VERSION = "20.12.0.0.0"  
TIMESTAMP = "2024/06/14 02:59:34"
```

```
[root@standalones hm]# cat version.txt  
AV_RELEASE_VERSION = "20.12.0.0.0"  
TIMESTAMP = "2024/06/14 02:59:34"  
[root@standalones hm]#
```

Dashboard:



4. Update Audit Vault Server (HA)

4.1.Process List Upgrade

- Update the standby Audit Vault Server.
- After you reboot the standby Audit Vault Server, ensure that it is up and running before updating the primary Audit Vault Server.
- Stop the audit trails on the primary Audit Vault Server.
- Update the primary Audit Vault Server.
- After you reboot the primary Audit Vault Server and confirm that it's running, no additional reboot is needed. It's fully functional at this point.

4.2.Update the Standby Audit Vault Server

4.2.1. Run the Pre-upgrade RPM

```
[root@avs005056a02204 ~]# sha256sum avdf-pre-upgrade-20.15.0.0.0.zip
a4ac68c04648c96d86afd3fd0207e2fb577d993619b5c64703b8203aeb6e8393  avdf-pre-upgrade-20.15.0.0.0.zip
[root@avs005056a02204 ~]#
[root@avs005056a02204 ~]# ll
total 384
-rw-----, 1 root root 5936 Jan 30 2003 anaconda-ks.cfg
-rw-----, 1 root root 37503 Aug 29 01:52 AVDF_20_15_readme.html
-rw-r--r--, 1 root root 170524 Sep 10 01:17 avdf-pre-upgrade-20.15.0.0.0-RELEASE.d.x86_64.rpm
-r--r--r--, 1 root root 158365 Sep 13 11:44 avdf-pre-upgrade-20.15.0.0.0.zip
-rw-r--r--, 1 root root 15 Jun 22 2018 dbfwboot.txt
-rw-----, 1 root root 5417 Jan 30 2003 original-ks.cfg
```

```

[root@avs005056a02204 ~]# rpm -i avdf-pre-upgrade-20.15.0.0.0-0_RELEASE.d.x86_64.rpm
active
Checking upgrade preconditions
avs-20.12.0.0.0-0_240614.0100.x86_64
Verifying certificate date: /usr/local/dbfw/etc/ca.crt
Verifying certificate date: /etc/pki/tls/certs/localhost_internal.crt
Verifying certificate date: /usr/local/dbfw/etc/cert.crt
Verifying certificate date: /usr/local/dbfw/etc/avs/avs_apex_client.crt
Verifying certificate date: /etc/pki/tls/certs/localhost.crt
Checking upgrade preconditions
Checking /var/lib/oracle/software
avs-20.12.0.0.0-0_240614.0100.x86_64
Verifying certificate date: /usr/local/dbfw/etc/ca.crt
Verifying certificate date: /etc/pki/tls/certs/localhost_internal.crt
Verifying certificate date: /usr/local/dbfw/etc/cert.crt
Verifying certificate date: /usr/local/dbfw/etc/avs/avs_apex_client.crt
Verifying certificate date: /etc/pki/tls/certs/localhost.crt
/usr/local/dbfw/bin/upgrade/oci-pki-check.rb: warning: argument of top-level return is ignored
Shutting down services.

AVDF::Installer::Upgrade::WarningPreconditions

Remove any schemas you have added to the database.

Additional schemas will cause the upgrade to fail.

SUCCESS:

The upgrade media can now be copied to '/var/dbfw/upgrade'.

The upgrade can then be started by running as root user:

    /usr/bin/avdf-upgrade

[root@avs005056a02204 ~]# █

```

4.2.2. Transfer the ISO File to the Appliance

```

[root@standalone avdf]# scp avdf-upgrade-20.15.0.0.0.iso support@192.168.112.120:/var/lib/oracle/software
Audit Vault Server 20.12.0.0
DO NOT CHANGE ANY CONFIGURATIONS IN Audit Vault Server APPLIANCE WITHOUT GUIDANCE FROM
ORACLE SUPPORT. ANY CHANGES SHOULD BE TRACEABLE TO APPROPRIATE SR REFERENCE.
support@192.168.112.120's password:
avdf-upgrade-20.15.0.0.0.iso 100% 16GB 106.9MB/s 02:35

```

4.2.3. Start the Update Script

```
[root@avs005056a02204 ~]# /usr/bin/avdf-upgrade
'/Images' already mounted, Unmounting...
Unmounted /dev/mapper/vg_root-lv images on /images
Please wait while validating SHA256 checksum for /var/dbfw/upgrade/avdf-upgrade-20.15.0.0.0.iso
Checksum validation successful for /var/dbfw/upgrade/avdf-upgrade-20.15.0.0.0.iso
Mounting /var/dbfw/upgrade/avdf-upgrade-20.15.0.0.0.iso on /images
mount: /images: WARNING: device write-protected, mounted read-only.
Successfully mounted /var/dbfw/upgrade/avdf-upgrade-20.15.0.0.0.iso on /images
/images/upgrade/lib/common.rb:1002: warning: mismatched indentations at 'rescue' with 'def' at 993
/images/upgrade/lib/chroot_upgrade.rb:337: warning: assigned but unused variable - ret

The following messages have important information about the upgrade process.

Power loss during upgrade may cause data loss. Do not power
off during upgrade.

Please review Note ID 2235931.1 for a current list of known issues.

The upgrade process is irreversible, please confirm 'y' to continue or 'n' to abort. [y/N]?
y
Verifying upgrade preconditions
avs-20.12.0.0.0-0_240614.0100.x86_64
Verifying certificate date: /usr/local/dbfw/etc/ca.crt
Verifying certificate date: /etc/pki/tls/certs/localhost_internal.crt
Verifying certificate date: /usr/local/dbfw/etc/cert.crt
Verifying certificate date: /usr/local/dbfw/etc/avs/avs_apex_client.crt
Verifying certificate date: /etc/pki/tls/certs/localhost.crt
mke2fs 1.45.6 (20-Mar-2020)
Discarding device blocks: done
Creating filesystem with 512000 4k blocks and 128000 inodes
Filesystem UUID: c950b1f3-3571-4a1c-a446-7eb1a6739454
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

/images/upgrade/aux//oci-pki-check.rb: warning: argument of top-level return is ignored
1/11: Mounting filesystems (1)
2/11: Cleaning yum configuration
3/11: Cleaning old packages and files
Warning: failed to remove kernel-tools-libs
4/11: Upgrading kernel
5/11: Upgrading system
6/11: Cleaning platform packages repo
7/11: Adding required platform packages
8/11: Cleaning AVDF packages repo
9/11: Installing AVDF packages
10/11: Setting boot title
11/11: Setting final system status
Reboot now to continue the upgrade process.
Unmounted /var/dbfw/upgrade/avdf-upgrade-20.15.0.0.0.iso on /images
```

4.2.4. Restart the Appliance

```
[root@avs005056a02204 ~]# reboot

Remote side unexpectedly closed network connection

Session stopped
- Press <Return> to exit tab
- Press R to restart session
- Press S to save terminal output to file
```

Checking background proses:

```
watch -n 5 "/opt/avdf/bin/privmigutl --status"
```

```
Every 5.0s: /opt/avdf/bin/privmigutl --status avs005056a02204: Tue Sep 23 03:04:36 2025

Migration set 'DEPENDENCIES' - started
Last migration 'Beginning installation, or upgrade, of Audit Vault Server dependencies' - started
The upgrade is in progress or was interrupted. Please wait until the upgrade completes or contact support.
```

/opt/avdf/bin/privmigutl --history

```
[root@avs005056a02204 ~]# /opt/avdf/bin/privmigutl --history
2018-06-22 12:13:21 Migration TOP:1, package_helper["-c", "/media/avdf-install/avs-bootstrap/yum.conf", "--repo", "avdf-install-bootstrap", "-i", "bootstrap-os"] (as root) - succeeded
2018-06-22 12:14:52 Migration TOP:2, install-common-os-precheck (as root) - succeeded
2018-06-22 12:14:52 Migration TOP:3, run-privileged-migrations["common-os"] (as root; retry permitted) - started
2018-06-22 12:14:52 Migration COMMON-05:1, noop (as root) - succeeded
2018-06-22 12:14:53 Migration COMMON-05:2, selinux.rb (as root) - succeeded
2018-06-22 12:14:53 Migration COMMON-05:3, stig-yum-config.rb (as root) - succeeded
2018-06-22 12:14:53 Migration COMMON-05:4, configure_systemd_service.rb["avdf-services.yaml"] (as root) - succeeded
2018-06-22 12:14:55 Migration TOP:3, run-privileged-migrations["common-os"] (as root; retry permitted) - succeeded
2018-06-22 12:14:55 Migration TOP:4, install-common-os-postcheck (as root) - succeeded
2018-06-22 12:15:08 Migration TOP:5, package_helper["-c", "/media/avdf-install/avs-bootstrap/yum.conf", "--repo", "avs-install-bootstrap", "-i", "bootstrap-avs-dependencies"] (as root) - succeeded
2018-06-22 12:15:08 Migration TOP:6, install-dependencies-precheck (as root) - succeeded
2018-06-22 12:15:08 Migration TOP:7, run-privileged-migrations["dependencies"] (as root; retry permitted) - started
2018-06-22 12:15:08 Migration DEPENDENCIES:1, update_conf_files.rb["--migration-target=dbfw.conf", "--mode=664", "--user=dbfw", "--group=dbfw"] (as root) - succeeded
2018-06-22 12:15:08 Migration DEPENDENCIES:2, configure_syslog (as root) - succeeded
2018-06-22 12:15:09 Migration DEPENDENCIES:3, lvm-adjust.rb["--config=lv-sanitisation"] (as root) - succeeded
2018-06-22 12:16:12 Migration DEPENDENCIES:4, import-network-settings (as root) - succeeded
2018-06-22 12:16:12 Migration DEPENDENCIES:5, configure_nics-config.rb (as root) - succeeded
2018-06-22 12:16:15 Migration DEPENDENCIES:6, configure_networking.rb["run-only-nic_detection, iptables, disable_devices, create_nics, modify_nics, hosts, hostname, sshd, apache, routing", "no-restart-httpd"] (as root) - succeeded
2018-06-22 12:16:15 Migration DEPENDENCIES:7, ocl-dependencies.rb (as root) - succeeded
2018-06-22 12:16:16 Migration DEPENDENCIES:8, package_helper["-c", "/media/avdf-install/avs-ol-dependencies/yum.conf", "--repo", "avs-ol-dependencies", "--clean"] (as root) - succeeded
2018-06-22 12:17:22 Migration DEPENDENCIES:9, package_helper["-c", "/media/avdf-install/avs-ol-dependencies/yum.conf", "--repo", "avs-ol-dependencies", "-g", "-i", "avs-ol-dependencies"] (as root) - succeeded
2018-06-22 12:17:22 Migration DEPENDENCIES:10, package_helper["-c", "/media/avdf-install/avs-dependencies/yum.conf", "--repo", "avs-dependencies", "--clean"] (as root) - succeeded
2018-06-22 12:17:28 Migration DEPENDENCIES:11, package_helper["-c", "/media/avdf-install/avs-dependencies/yum.conf", "--repo", "avs-dependencies", "-i", "ruby"] (as root) - succeeded
2018-06-22 12:17:29 Migration DEPENDENCIES:12, configure_systemd_service.rb["avs-os-services.yaml"] (as root) - succeeded
2018-06-22 12:17:30 Migration DEPENDENCIES:13, package_helper["-c", "/media/avdf-install/avs-dependencies/yum.conf", "--repo", "avs-dependencies", "--clean"] (as root) - succeeded
2018-06-22 12:31:08 Migration DEPENDENCIES:14, package_helper["-c", "/media/avdf-install/avs-dependencies/yum.conf", "--repo", "avs-dependencies", "-g", "-i", "avs-dependencies, common-dependencies"] (as root) - succeeded
2018-06-22 12:31:09 Migration DEPENDENCIES:15, check-fstab-latest.rb (as root) - succeeded
2018-06-22 12:31:11 Migration DEPENDENCIES:16, secondary_nics.rb (as root) - succeeded
2018-06-22 12:31:22 Migration DEPENDENCIES:17, configure_networking.rb["run-only-disable_devices, create_nics, modify_nics"] (as root) - succeeded
2018-06-22 12:31:31 Migration DEPENDENCIES:18, change_shm_permissions.py (as root) - succeeded
2018-06-22 12:31:31 Migration DEPENDENCIES:19, update_sysctl_conf_hugepages.py (as root) - succeeded
2018-06-22 12:31:34 Migration DEPENDENCIES:20, stop_dbfw_listener (as root) - succeeded
2018-06-22 12:31:35 Migration DEPENDENCIES:21, upgrade_start_asm_db.py (as root) - succeeded
2018-06-22 12:31:35 Migration DEPENDENCIES:22, update_jobs["--install=upgrade", "--avs", "disable"] (as root) - succeeded
2018-06-22 12:31:35 Migration DEPENDENCIES:23, create_grid.sh (as root) - succeeded
2018-06-22 12:53:43 Migration DEPENDENCIES:24, create_database.sh (as root) - succeeded
2018-06-22 12:53:45 Migration DEPENDENCIES:25, remove_apex_devenv.py (as oracle) - succeeded
2018-06-22 12:53:45 Migration DEPENDENCIES:26, upgrade_oracle_restart (as root) - succeeded
2018-06-22 12:53:46 Migration DEPENDENCIES:27, disable_database_vault_new (as root) - succeeded
2018-06-22 12:53:46 Migration DEPENDENCIES:28, upgrade_dbfwdb (as root) - succeeded
2018-06-22 13:22:19 Migration DEPENDENCIES:29, apply_grid_r_u_update.py (as root) - succeeded
2018-06-22 13:22:19 Migration DEPENDENCIES:30, apply_db_r_u_update.py (as oracle) - succeeded
```

tail -f /var/log/messages

```
[root@avs005056a02204 ~]# tail -f /var/log/messages
Sep 23 03:00:01 avs005056a02204 systemd[1]: systat-collect.service: Succeeded.
Sep 23 03:00:01 avs005056a02204 systemd[1]: Started system activity accounting tool.
Sep 23 03:03:58 avs005056a02204 systemd[1]: Reloading.
Sep 23 03:04:06 avs005056a02204 systemd[1]: proc-sys-fs-binfmt_misc.automount: Got automount request for /proc/sys/fs/binfmt_misc, triggered by 12746 (jexec).
Sep 23 03:04:06 avs005056a02204 systemd[1]: Mounting Arbitrary Executable File Formats File System...
Sep 23 03:04:07 avs005056a02204 systemd[1]: Mounted Arbitrary Executable File Formats File System.
Sep 23 03:04:57 avs005056a02204 sshd[13666]: rexec line 19: Deprecated option UsePrivilegeSeparation
Sep 23 03:05:04 avs005056a02204 systemd[1]: Started Session 7 of user support.
Sep 23 03:05:04 avs005056a02204 systemd-logind[1293]: New session 7 of user support.
Sep 23 03:05:09 avs005056a02204 su: (to root) support on pts/2
Sep 23 03:05:32 avs005056a02204 sshd[14214]: rexec line 19: Deprecated option UsePrivilegeSeparation
Sep 23 03:05:39 avs005056a02204 sshd[14242]: rexec line 19: Deprecated option UsePrivilegeSeparation
Sep 23 03:05:38 avs005056a02204 systemd[1]: Started Session 8 of user support.
Sep 23 03:05:39 avs005056a02204 systemd-logind[1293]: New session 8 of user support.
Sep 23 03:05:39 avs005056a02204 systemd[1]: Started Session 9 of user support.
Sep 23 03:05:39 avs005056a02204 systemd-logind[1293]: New session 9 of user support.
Sep 23 03:05:43 avs005056a02204 su: (to root) support on pts/3
Sep 23 03:08:34 avs005056a02204 systemd[1]: Starting Cleanup of Temporary Directories...
Sep 23 03:08:35 avs005056a02204 systemd[1]: systemd-tmpfiles-clean.service: Succeeded.
Sep 23 03:08:35 avs005056a02204 systemd[1]: Started Cleanup of Temporary Directories.
Sep 23 03:10:00 avs005056a02204 systemd[1]: Starting system activity accounting tool...
Sep 23 03:10:00 avs005056a02204 systemd[1]: systat-collect.service: Succeeded.
Sep 23 03:10:00 avs005056a02204 systemd[1]: Started system activity accounting tool.
Sep 23 03:10:16 avs005056a02204 run-privileged-migrations[3558]: com.oracle.dbfw.privilegedMigration AUDIT - 00F-11004: Migration stanza result: success: package_helper["-c", "/media/avdf-install/avs-dependencies", "-g", "-i", "avs-dependencies, common-dependencies"] (as root) (no changes applied)
Sep 23 03:10:16 avs005056a02204 run-privileged-migrations[3558]: com.oracle.dbfw.privilegedMigration AUDIT - 00F-11003: Migration stanza invocation: check-fstab-latest.rb (as root)
Sep 23 03:10:17 avs005056a02204 run-privileged-migrations[3558]: com.oracle.dbfw.privilegedMigration AUDIT - 00F-11004: Migration stanza result: success: check-fstab-latest.rb (as root) (no changes applied)
Sep 23 03:10:17 avs005056a02204 run-privileged-migrations[3558]: com.oracle.dbfw.privilegedMigration AUDIT - 00F-11003: Migration stanza invocation: secondary_nics.rb (as root)
Sep 23 03:10:18 avs005056a02204 run-privileged-migrations[3558]: com.oracle.dbfw.privilegedMigration AUDIT - 00F-11004: Migration stanza result: success: secondary_nics.rb (as root) (no changes applied)
Sep 23 03:10:18 avs005056a02204 run-privileged-migrations[3558]: com.oracle.dbfw.privilegedMigration AUDIT - 00F-11003: Migration stanza invocation: configure_networking.rb["run-only-disable_devices, create_nics, modify_nics, listener_oracle, sqlnet_oracle, tnsnames_oracle", "no-regen-agent", "no-restart-dbfwlistener", "no-restart-controller"] (as root)
Sep 23 03:10:22 avs005056a02204 run-privileged-migrations[3558]: com.oracle.dbfw.privilegedMigration AUDIT - 00F-11004: Migration stanza result: success: configure_networking.rb["run-only-disable_devices, create_nics, modify_nics, listener_oracle, sqlnet_oracle, tnsnames_oracle", "no-regen-agent", "no-restart-dbfwlistener", "no-restart-controller"] (as root) (applied change)
Sep 23 03:10:22 avs005056a02204 run-privileged-migrations[3558]: com.oracle.dbfw.privilegedMigration AUDIT - 00F-11003: Migration stanza invocation: change_shm_permissions.py (as root)
Sep 23 03:10:22 avs005056a02204 root: Starting execution of Oracle Clusterware init.ohasd
```

tail -f /var/log/debug

Database firewall still can use:

```
[oracle@standalone ~]$ !221
sqlplus avdfuser/12345@192.168.112.178:15221/pdb1

SQL*Plus: Release 19.0.0.0.0 - Production on Tue Sep 23 14:21:47 2025
Version 19.28.0.0.0

Copyright (c) 1982, 2025, Oracle. All rights reserved.

Last Successful login time: Mon Sep 22 2025 14:18:05 +07:00

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.28.0.0.0

SQL>
SQL> exit
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.28.0.0.0
[oracle@standalone ~]$
```

Finish upgrade:

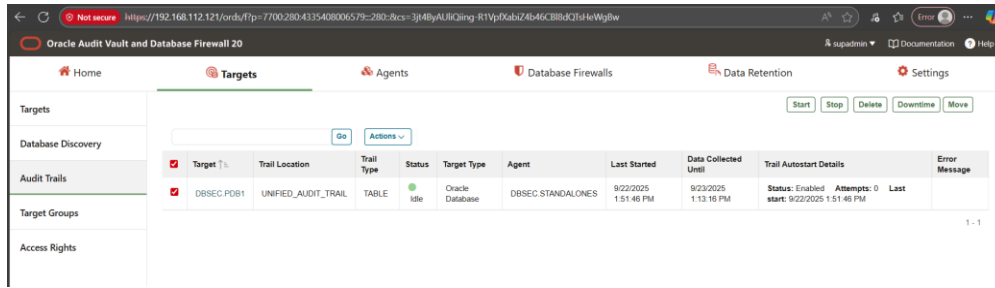
```
Every 5.0s: /opt/avdf/bin/privmigutl --status

Migration set 'TOP' - succeeded
Last migration 'Switching to full runtime' - succeeded
No further action needed.
```

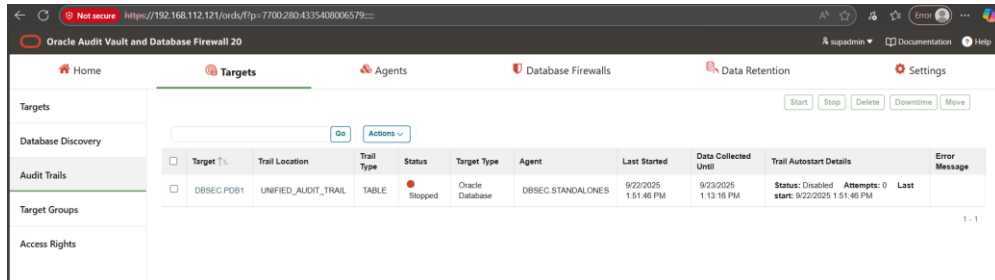
```
login as: support
Pre-authentication banner message from server:
| Audit Vault Server 20.15.0.0.0
|
| DO NOT CHANGE ANY CONFIGURATIONS IN Audit Vault Server APPLIANCE WITHOUT GUID
| ANCE FROM
| ORACLE SUPPORT. ANY CHANGES SHOULD BE TRACEABLE TO APPROPRIATE SR REFERENCE.
End of banner message from server
support@192.168.112.120's password:
```

4.3.Stop All Audit Trails

- Log into the Audit Vault Server console as an administrator.
- Click the Targets tab.
- Click Audit Trails in the left navigation menu.
- Select all audit trails.



- Click Stop.



4.4.Update the Primary Audit Vault Server

4.4.1. Run the Pre-upgrade RPM

```
[root@avs005056a02dd1 ~]# sha256sum avdf-pre-upgrade-20.15.0.0.zip
a4ac68c04648c96d86afd3fd0207e2fb577d993619b5c64703b8203aeb6e8393 avdf-pre-upgrade-20.15.0.0.zip
[root@avs005056a02dd1 ~]#
[root@avs005056a02dd1 ~]# unzip avdf-pre-upgrade-20.15.0.0.zip
Archive: avdf-pre-upgrade-20.15.0.0.zip
  inflating: avdf-pre-upgrade-20.15.0.0-0_RELEASE.d.x86_64.rpm
[root@avs005056a02dd1 ~]# ll
total 352
-rw-r--r-- 1 root root 5936 Jan 30 2003 anaconda-ks.cfg
-rw-r--r-- 1 root root 170524 Sep 10 01:17 avdf-pre-upgrade-20.15.0.0-0_RELEASE.d.x86_64.rpm
-r--r--r-- 1 support support 158365 Sep 23 06:11 avdf-pre-upgrade-20.15.0.0.zip
-r--r--r-- 1 root root 8036 Sep 16 09:43 avs-source-20.14-replication.rpm
-rw-r--r-- 1 root root 15 Jun 22 2018 dbfwboot.txt
-rw-r--r-- 1 root root 5417 Jan 30 2003 original-ks.cfg
[root@avs005056a02dd1 ~]#
```

```
[root@avs005056a02dd1 ~]# rpm -i avdf-pre-upgrade-20.15.0.0.0-0_RELEASE.d.x86_64.rpm
active
Checking upgrade preconditions
avs-20.12.0.0.0-0_240614.0100.x86_64
Verifying certificate date: /usr/local/dbfw/etc/ca.crt
Verifying certificate date: /etc/pki/tls/certs/localhost_internal.crt
Verifying certificate date: /usr/local/dbfw/etc/cert.crt
Verifying certificate date: /usr/local/dbfw/etc/avs/avs_apex_client.crt
Verifying certificate date: /etc/pki/tls/certs/localhost.crt
Oracle Interim Patch Installer version 12.2.0.1.42
Copyright (c) 2025, Oracle Corporation. All rights reserved.

Oracle Home      : /var/lib/oracle/grid
Central Inventory : /var/lib/oracle/orainventory
   from          : /var/lib/oracle/grid/orainst.loc
OPatch version   : 12.2.0.1.42
OUI version      : 12.2.0.7.0
Log file location : /var/lib/oracle/grid/cfgtoollogs/patch/patch2025-09-23_06-14-26AM_1.log

Invoking utility "cleanup"
OPatch will clean up 'restore.sh,make.txt' files and 'scratch,backup' directories.
You will be still able to rollback patches after this cleanup.
Do you want to proceed? [y|n]
Y (auto-answered by -silent)
User Responded with: Y

Backup area for restore has been cleaned up. For a complete list of files/directories
deleted, Please refer log file.

OPatch succeeded.
Oracle Interim Patch Installer version 12.2.0.1.42
Copyright (c) 2025, Oracle Corporation. All rights reserved.

Oracle Home      : /var/lib/oracle/dbfw
Central Inventory : /var/lib/oracle/orainventory
   from          : /var/lib/oracle/dbfw/orainst.loc
OPatch version   : 12.2.0.1.42
OUI version      : 12.2.0.7.0
Log file location : /var/lib/oracle/dbfw/cfgtoollogs/patch/patch2025-09-23_06-15-12AM_1.log

Invoking utility "cleanup"
OPatch will clean up 'restore.sh,make.txt' files and 'scratch,backup' directories.
You will be still able to rollback patches after this cleanup.
Do you want to proceed? [y|n]
Y (auto-answered by -silent)
User Responded with: Y

Backup area for restore has been cleaned up. For a complete list of files/directories
deleted, Please refer log file.

OPatch succeeded.
Checking upgrade preconditions
```

```
OPatch succeeded.
Checking upgrade preconditions
Checking /var/lib/oracle/software
avs-20.12.0.0.0-0_240614.0100.x86_64
Verifying certificate date: /usr/local/dbfw/etc/ca.crt
Verifying certificate date: /etc/pki/tls/certs/localhost_internal.crt
Verifying certificate date: /usr/local/dbfw/etc/cert.crt
Verifying certificate date: /usr/local/dbfw/etc/avs/avs_apex_client.crt
Verifying certificate date: /etc/pki/tls/certs/localhost.crt
Pre Upgrade version : 20.15.0.0.0
Sent Agent/HM Prerequisites checks message successfully. Execute /usr/local/dbfw/bin/upgrade/pre_upgrade_download_agent_validation_status.py script to see validation checks status.
Downloaded prerequisites check status reports successfully at /opt/avdf/report
WARNING: Before proceeding with upgrade, it is recommended to resolve all the agent related pre-upgrade warnings and errors in /opt/avdf/report/agent_prereq_checks_success_report.txt and /opt/avdf/report/agent_prereq_checks_failure_report.txt respectively.
Oracle Interim Patch Installer version 12.2.0.1.42
Copyright (c) 2025, Oracle Corporation. All rights reserved.

Oracle Home      : /var/lib/oracle/grid
Central Inventory : /var/lib/oracle/orainventory
   from          : /var/lib/oracle/grid/orainst.loc
OPatch version   : 12.2.0.1.42
OUI version      : 12.2.0.7.0
Log file location : /var/lib/oracle/grid/cfgtoollogs/patch/patch2025-09-23_06-17-35AM_1.log

Invoking utility "cleanup"
OPatch will clean up 'restore.sh,make.txt' files and 'scratch,backup' directories.
You will be still able to rollback patches after this cleanup.
Do you want to proceed? [y|n]
Y (auto-answered by -silent)
User Responded with: Y

Backup area for restore has been cleaned up. For a complete list of files/directories
deleted, Please refer log file.

OPatch succeeded.
Oracle Interim Patch Installer version 12.2.0.1.42
Copyright (c) 2025, Oracle Corporation. All rights reserved.

Oracle Home      : /var/lib/oracle/dbfw
Central Inventory : /var/lib/oracle/orainventory
   from          : /var/lib/oracle/dbfw/orainst.loc
OPatch version   : 12.2.0.1.42
OUI version      : 12.2.0.7.0
Log file location : /var/lib/oracle/dbfw/cfgtoollogs/patch/patch2025-09-23_06-18-13AM_1.log

Invoking utility "cleanup"
OPatch will clean up 'restore.sh,make.txt' files and 'scratch,backup' directories.
You will be still able to rollback patches after this cleanup.
Do you want to proceed? [y|n]
Y (auto-answered by -silent)
```

```

Invoking utility "cleanup"
OPatch will clean up 'restore.sh,make.txt' files and 'scratch,backup' directories.
You will be still able to rollback patches after this cleanup.
Do you want to proceed? [y|n]
Y (auto-answered by -silent)
User Responded with: Y

Backup area for restore has been cleaned up. For a complete list of files/directories
deleted, Please refer log file.

OPatch succeeded.
/usr/local/dbfw/bin/upgrade/oci-pki-check.rb: warning: argument of top-level return is ignored
Shutting down services.

AVDF::Installer::Upgrade::WarningPreconditions

Remove any schemas you have added to the database.

Additional schemas will cause the upgrade to fail.

SUCCESS:

The upgrade media can now be copied to '/var/dbfw/upgrade'.

The upgrade can then be started by running as root user:

/usr/bin/avdf-upgrade
[root@avs005056a02dd1 ~]# █

```

4.4.2. Transfer the ISO File to the Appliance

```

[root@standalone avdf]# ll
total 16999136
-rw-r--r--. 1 root root 17407113216 Sep 22 22:23 avdf-upgrade-20.15.0.0.0.iso
[root@standalone avdf]# scp avdf-upgrade-20.15.0.0.0.iso support@192.168.112.121:/var/lib/oracle/software
Audit Vault Server 20.12.0.0.0

DO NOT CHANGE ANY CONFIGURATIONS IN Audit Vault Server APPLIANCE WITHOUT GUIDANCE FROM
ORACLE SUPPORT. ANY CHANGES SHOULD BE TRACEABLE TO APPROPRIATE SR REFERENCE.
support@192.168.112.121's password:
avdf-upgrade-20.15.0.0.0.iso
100% 16GB 104.3MB/s 02:39
[root@standalone avdf]# █

[root@avs005056a02dd1 software]# ll
total 16999140
-rw-r--r--. 1 support support 17407113216 Sep 23 06:27 avdf-upgrade-20.15.0.0.0.iso
[root@avs005056a02dd1 software]# mv avdf-upgrade-20.15.0.0.0.iso /var/dbfw/upgrade
[root@avs005056a02dd1 software]#
[root@avs005056a02dd1 software]# ls -ltr /var/dbfw/upgrade
total 16999156
drwx----- 2 root root 16384 Jan 30 2003 lost+found
-rw-r--r--. 1 support support 17407113216 Sep 23 06:27 avdf-upgrade-20.15.0.0.0.iso
[root@avs005056a02dd1 software]#

```

4.4.3. Start the Update Script

```

[root@avs005056a02dd1 ~]# /usr/bin/avdf-upgrade
'/Images' already mounted, Unmounting...
Unmounted /dev/mapper/vg_root-lv_images on /images
Please wait while validating SHA256 checksum for /var/dbfw/upgrade/avdf-upgrade-20.15.0.0.0.iso
Checksum validation successful for /var/dbfw/upgrade/avdf-upgrade-20.15.0.0.0.iso
Mounting /var/dbfw/upgrade/avdf-upgrade-20.15.0.0.0.iso on /images
mount: /images: WARNING: device write-protected, mounted read-only.
Successfully mounted /var/dbfw/upgrade/avdf-upgrade-20.15.0.0.0.iso on /images
/images/upgrade/lib/common.rb:1002: warning: mismatched indentations at 'rescue' with 'def' at 993
/images/upgrade/lib/chroot_upgrade.rb:337: warning: assigned but unused variable - ret

The following messages have important information about the upgrade process.

Power loss during upgrade may cause data loss. Do not power
off during upgrade.

Please review Note ID 2235931.1 for a current list of known issues.

The upgrade process is irreversible, please confirm 'y' to continue or 'n' to abort. [y/N]?
y
Verifying upgrade preconditions
avs-20.12.0.0.0-0_240614.0100.x86_64
Verifying certificate date: /usr/local/dbfw/etc/ca.crt
Verifying certificate date: /etc/pki/tls/certs/localhost_internal.crt
Verifying certificate date: /usr/local/dbfw/etc/cert.crt
Verifying certificate date: /usr/local/dbfw/etc/avs/avs_apex_client.crt
Verifying certificate date: /etc/pki/tls/certs/localhost.crt
mke2fs 1.45.6 (20-Mar-2020)
Discarding device blocks: done
Creating filesystem with 512000 4k blocks and 128000 inodes
Filesystem UUID: 0de64c81-5701-42df-99ce-236683931113
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

/images/upgrade/aux//oci-pki-check.rb: warning: argument of top-level return is ignored
1/11: Mounting filesystems (1)
2/11: Cleaning yum configuration
3/11: Cleaning old packages and files
Warning: failed to remove kernel-tools-libs
4/11: Upgrading kernel
5/11: Upgrading system
6/11: Cleaning platform packages repo
7/11: Adding required platform packages
8/11: Cleaning AVDF packages repo
9/11: Installing AVDF packages
10/11: Setting boot title
11/11: Setting final system status
Reboot now to continue the upgrade process.
Unmounted /var/dbfw/upgrade/avdf-upgrade-20.15.0.0.0.iso on /images
[root@avs005056a02dd1 ~]#

```

4.4.4. Restart the Appliance

```

[root@avs005056a02dd1 ~]# reboot

Remote side unexpectedly closed network connection

Session stopped
- Press <Return> to exit tab
- Press R to restart session
- Press S to save terminal output to file

```

Checking background proeses:

watch -n 5 "/opt/avdf/bin/privmigutl --status"

```
Every 5.0s: /opt/avdf/bin/privmigutl --status av005056a02dd1: Tue Sep 23 06:48:24 2025

Migration set 'DEPENDENCIES' - started
Last migration 'Beginning installation, or upgrade, of Audit Vault Server dependencies' - started
The upgrade is in progress or was interrupted. Please wait until the upgrade completes or contact support.
```

/opt/avdf/bin/privmigutl --history

```
[root@av005056a02dd1 ~]# /opt/avdf/bin/privmigutl --history
2018-06-22 12:13:45 Migration TOP:1, package_helper["-c", "/media/avdf-install/avs-bootstrap/yum.conf", "--repo", "avdf-install-bootstrap", "--l", "bootstrap-os"] (as root) - succeeded
2018-06-22 12:15:09 Migration TOP:2, install-common-os-precheck (as root) - succeeded
2018-06-22 12:15:09 Migration TOP:3, run-privileged-migrations["common-os"] (as root; retry permitted) - started
2018-06-22 12:15:09 Migration COMMON-OS:1, noop (as root) - succeeded
2018-06-22 12:15:09 Migration DEPENDENCIES:2, configure-syslog (as root) - succeeded
2018-06-22 12:15:10 Migration COMMON-OS:3, stig-yum-config.rb (as root) - succeeded
2018-06-22 12:15:12 Migration COMMON-OS:4, configure_systemd_service.rb["avdf-services.yaml"] (as root) - succeeded
2018-06-22 12:15:12 Migration TOP:3, run-privileged-migrations["common-os"] (as root; retry permitted) - succeeded
2018-06-22 12:15:12 Migration TOP:4, install-common-os-postcheck (as root) - succeeded
2018-06-22 12:17:04 Migration DEPENDENCIES:5, preserve-nic-config.rb (as root) - succeeded
2018-06-22 12:15:24 Migration TOP:5, package_helper["-c", "/media/avdf-install/avs-bootstrap/yum.conf", "--repo", "avs-install-bootstrap", "--l", "bootstrap-avs-dependencies"] (as root) - succeeded
2018-06-22 12:15:24 Migration TOP:6, install-dependencies-precheck (as root) - succeeded
2018-06-22 12:15:24 Migration TOP:7, run-privileged-migrations["dependencies"] (as root; retry permitted) - started
2018-06-22 12:15:25 Migration DEPENDENCIES:1, update_conf_files.rb["--migration-target=dbfw.conf", "--mode=64", "--user=dbfw", "--group=dbfw"] (as root) - succeeded
2018-06-22 12:15:25 Migration DEPENDENCIES:3, lvm-adjust.rb["--config=lvm-sanitisation"] (as root) - succeeded
2018-06-22 12:17:01 Migration DEPENDENCIES:4, import-network-settings (as root) - succeeded
2018-06-22 12:17:04 Migration DEPENDENCIES:6, configure_networking.rb["run-only=nic_detection,iptables,disable_devices,create_nics,modify_nics,hosts,hostname,ssh,apache_routing", "no-restart-httpd"] (as root) - succeeded
2018-06-22 12:17:04 Migration DEPENDENCIES:7, oci-dependencies.rb (as root) - succeeded
2018-06-22 12:17:05 Migration DEPENDENCIES:8, package_helper["-c", "/media/avdf-install/avs-ol-dependencies/yum.conf", "--repo", "avs-ol-dependencies", "--clean"] (as root) - succeeded
2018-06-22 12:18:18 Migration DEPENDENCIES:9, package_helper["-c", "/media/avdf-install/avs-ol-dependencies/yum.conf", "--repo", "avs-ol-dependencies", "-g", "-l", "avs-ol-dependencies"] (as root) - succeeded
2018-06-22 12:18:19 Migration DEPENDENCIES:10, package_helper["-c", "/media/avdf-install/avs-dependencies/yum.conf", "--repo", "avs-dependencies", "--clean"] (as root) - succeeded
2018-06-22 12:18:20 Migration DEPENDENCIES:11, package_helper["-c", "/media/avdf-install/avs-dependencies/yum.conf", "--repo", "avs-dependencies", "-l", "ruby"] (as root) - succeeded
2018-06-22 12:18:21 Migration DEPENDENCIES:12, configure_systemd_service.rb["avs-os-services.yaml"] (as root) - succeeded
2018-06-22 12:18:21 Migration DEPENDENCIES:13, package_helper["-c", "/media/avdf-install/avs-dependencies/yum.conf", "--repo", "avs-dependencies", "--clean"] (as root) - succeeded
2018-06-22 12:32:11 Migration DEPENDENCIES:14, package_helper["-c", "/media/avdf-install/avs-dependencies/yum.conf", "--repo", "avs-dependencies", "-g", "-l", "avs-dependencies-common-dependencies"] (as root) - succeeded
2018-06-22 12:32:12 Migration DEPENDENCIES:15, check-fstab-latest.rb (as root) - succeeded
2018-06-22 12:32:15 Migration DEPENDENCIES:16, secondary-nics.rb (as root) - succeeded
2018-06-22 12:32:17 Migration DEPENDENCIES:17, configure_networking.rb["run-only=disable_devices,create_nics,modify_nics"] (as root) - succeeded
2018-06-22 12:32:22 Migration DEPENDENCIES:18, change_shm_permissions.py (as root) - succeeded
2018-06-22 12:32:22 Migration DEPENDENCIES:19, update_sysctl_conf_hugepages.py (as root) - succeeded
2018-06-22 12:32:22 Migration DEPENDENCIES:20, stop_dbfw_listener (as root) - succeeded
2018-06-22 12:32:26 Migration DEPENDENCIES:21, upgrade_start_asm_db.py (as root) - succeeded
2018-06-22 12:32:27 Migration DEPENDENCIES:22, update_jobs["--install-upgrade", "--avs", "disable"] (as root) - succeeded
2018-06-22 12:39:25 Migration DEPENDENCIES:23, create_grid.sh (as root) - succeeded
2018-06-22 12:54:29 Migration DEPENDENCIES:24, create_database.sh (as root) - succeeded
2018-06-22 12:54:31 Migration DEPENDENCIES:25, remove_apex_devenv.py (as oracle) - succeeded
2018-06-22 12:54:31 Migration DEPENDENCIES:26, upgrade_oracle_restart (as root) - succeeded
2018-06-22 12:54:32 Migration DEPENDENCIES:27, disable-database-vault-tls (as root) - succeeded
2018-06-22 12:54:32 Migration DEPENDENCIES:28, upgrade_dbfwdb (as root) - succeeded
2018-06-22 13:23:45 Migration DEPENDENCIES:29, apply_grid_run_update.py (as root) - succeeded
2018-06-22 13:23:45 Migration DEPENDENCIES:30, apply_db_run_update.py (as oracle) - succeeded
2018-06-22 13:27:17 Migration DEPENDENCIES:31, upgrade_apex (as oracle) - succeeded
```

tail -f /var/log/messages

```
[root@av005056a02dd1 ~]# tail -f /var/log/messages
Sep 23 06:48:51 av005056a02dd1 systemd[1]: Started Session 8 of user support.
Sep 23 06:48:51 av005056a02dd1 systemd-logind[1207]: New session 8 of user support.
Sep 23 06:48:54 av005056a02dd1 su: (to root) support on pts/2
Sep 23 06:49:32 av005056a02dd1 sshd[6936]: rexec line 19: Deprecated option UsePrivilegeSeparation
Sep 23 06:49:35 av005056a02dd1 sshd[6971]: rexec line 19: Deprecated option UsePrivilegeSeparation
Sep 23 06:49:35 av005056a02dd1 systemd-logind[1207]: New session 9 of user support.
Sep 23 06:49:36 av005056a02dd1 su: (to root) support on pts/3
Sep 23 06:50:04 av005056a02dd1 systemd[1]: Starting system activity accounting tool...
Sep 23 06:50:04 av005056a02dd1 systemd[1]: sysstat-collect.service: Succeeded.
Sep 23 06:50:05 av005056a02dd1 systemd[1]: Started system activity accounting tool.
Sep 23 06:50:16 av005056a02dd1 systemd[423]: Starting Mark boot as successful...
Sep 23 06:50:16 av005056a02dd1 systemd[5423]: Started Mark boot as successful.
Sep 23 06:50:46 av005056a02dd1 systemd[1]: Reloading.
Sep 23 06:50:47 av005056a02dd1 systemd[1]: proc-sys-fs-binfmt_misc.automount: Got automount request for /proc/sys/fs/binfmt_misc, triggered by 13984 (jexec)
Sep 23 06:50:47 av005056a02dd1 systemd[1]: Mounting Arbitrary Executable File Formats File System...
Sep 23 06:50:47 av005056a02dd1 systemd[1]: Mounted Arbitrary Executable File Formats File System.
Sep 23 07:00:01 av005056a02dd1 systemd[1]: Starting system activity accounting tool...
Sep 23 07:00:01 av005056a02dd1 systemd[1]: sysstat-collect.service: Succeeded.
Sep 23 07:00:01 av005056a02dd1 systemd[1]: Started system activity accounting tool.
Sep 23 07:01:00 av005056a02dd1 systemd[1]: Starting Cleanup of Temporary Directories...
Sep 23 07:01:00 av005056a02dd1 systemd[1]: systemd-tmpfiles-clean.service: Succeeded.
Sep 23 07:01:01 av005056a02dd1 systemd[1]: Started Cleanup of Temporary Directories.
Sep 23 07:03:18 av005056a02dd1 run-privileged-migrations[3476]: com.oracle.dbfw.privilegedMigration AUDIT - 00F-11004: Migration stanza result: success: package_helper["-c", "/media/avdf-install/avs-dependencies/yum.conf", "--repo", "avs-dependencies", "-g", "-l", "avs-dependencies-common-dependencies"] (as root) (no changes applied)
Sep 23 07:03:19 av005056a02dd1 run-privileged-migrations[3476]: com.oracle.dbfw.privilegedMigration AUDIT - 00F-11003: Migration stanza invocation: check-fstab-latest.rb (as root) (no changes applied)
Sep 23 07:03:19 av005056a02dd1 run-privileged-migrations[3476]: com.oracle.dbfw.privilegedMigration AUDIT - 00F-11004: Migration stanza result: success: check-fstab-latest.rb (as root) (no changes applied)
Sep 23 07:03:20 av005056a02dd1 run-privileged-migrations[3476]: com.oracle.dbfw.privilegedMigration AUDIT - 00F-11003: Migration stanza invocation: secondary-nics.rb (as root) (no changes applied)
Sep 23 07:03:20 av005056a02dd1 run-privileged-migrations[3476]: com.oracle.dbfw.privilegedMigration AUDIT - 00F-11004: Migration stanza result: success: secondary-nics.rb (as root) (no changes applied)
Sep 23 07:03:20 av005056a02dd1 run-privileged-migrations[3476]: com.oracle.dbfw.privilegedMigration AUDIT - 00F-11003: Migration stanza invocation: configure_networking.rb["run-only=disable_devices,create_nics,modify_nics,listener_ora,sqlnet_ora,tnsnames.ora", "no-regen-agent", "no-restart-dbfwlistener", "no-restart-controller"] (as root) (no changes applied)
Sep 23 07:03:20 av005056a02dd1 run-privileged-migrations[3476]: com.oracle.dbfw.privilegedMigration AUDIT - 00F-11004: Migration stanza result: success: configure_networking.rb["run-only=disable_devices,create_nics,modify_nics,listener_ora,sqlnet_ora,tnsnames.ora", "no-regen-agent", "no-restart-dbfwlistener", "no-restart-controller"] (as root) (no changes applied)
Sep 23 07:03:20 av005056a02dd1 run-privileged-migrations[3476]: com.oracle.dbfw.privilegedMigration AUDIT - 00F-11003: Migration stanza invocation: change_shm_permissions.py (as root) (no changes applied)
Sep 23 07:03:23 av005056a02dd1 root: Starting execution of Oracle Clusterware init.ohasd
Sep 23 07:03:24 av005056a02dd1 su: (to grid) root on name
Sep 23 07:03:24 av005056a02dd1 systemd[1]: Stopping Oracle High Availability Services...
Sep 23 07:03:24 av005056a02dd1 init.ohasd[21145]: Found cgroups on /sys/fs/cgroup/systemd, legacy hierarchy
Sep 23 07:03:24 av005056a02dd1 systemd[1]: Started Session c23 of user grid.
Sep 23 07:03:25 av005056a02dd1 cscope: /etc/init.d/init.ohasd: Found cgroups on /sys/fs/cgroup/systemd, legacy hierarchy
Sep 23 07:03:25 av005056a02dd1 su: (to grid) root on name
Sep 23 07:03:25 av005056a02dd1 init.ohasd[21145]: /etc/init.d/init.ohasd: init.ohasd stop called while not running, taking no action
Sep 23 07:03:25 av005056a02dd1 systemd[1]: session-c23.scope: Succeeded.
Sep 23 07:03:25 av005056a02dd1 init.ohasd[21171]: init.ohasd start called while not running, taking no action
Sep 23 07:03:25 av005056a02dd1 systemd[1]: Started Session c24 of user grid.
Sep 23 07:03:25 av005056a02dd1 init.ohasd[21171]: /etc/init.d/init.ohasd: init.ohasd stop called while not running, taking no action
```

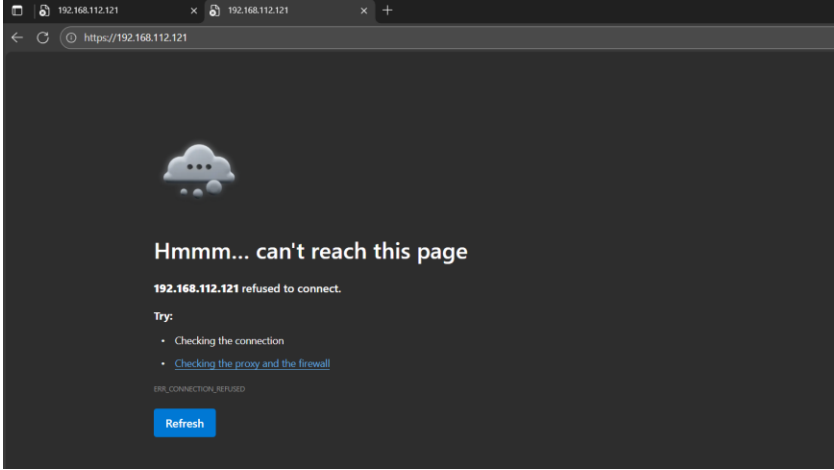
tail -f /var/log/debug


```

root@aws055e56a2d01:~# ./fvar/job/deb
Sdp 23 06:47:28 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: Install 7 Packages
Sdp 23 06:48:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: total size: 9.8 G
Sdp 23 06:48:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: installed size: 9.5 G
Sdp 23 06:48:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: Downloading Packages
Sdp 23 06:49:26 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: Running transaction check
Sdp 23 06:49:26 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: Transaction check succeeded.
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: Running transaction tests
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 2130 reads, 352989602 total bytes in 0.814219 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 64377 reads, 2189261684 total bytes in 17.828611 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 13 read, 187648 total bytes in 0.000008 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 6006 reads, 19685950 total bytes in 1.609135 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 120859 reads, 3960074748 total bytes in 15.794259 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 5364 reads, 306597028 total bytes in 1.265557 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 6 reads, 17488 total bytes in 0.000012 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 6 reads, 2984 total bytes in 0.000008 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 6 reads, 12820 total bytes in 0.000008 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 6 reads, 107904 total bytes in 0.000008 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 6 reads, 1992 total bytes in 0.000008 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 6 reads, 2312 total bytes in 0.000008 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: Transaction test succeeded.
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: Running transaction
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 1730 reads, 5579028 total bytes in 0.015152 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 92475 reads, 302990602 total bytes in 14.081898 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 64377 reads, 2189261684 total bytes in 15.457461 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 13 reads, 187648 total bytes in 0.000008 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 196859 reads, 19685950 total bytes in 1.609135 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 120859 reads, 3960074748 total bytes in 15.794259 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 5364 reads, 306597028 total bytes in 1.265557 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: Preparing...
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 2 writes, 41424 total bytes in 0.000044 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 2 writes, 59096 total bytes in 0.000072 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 2565 writes, 8402992 total bytes in 0.067266 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 24 writes, 1058304 total bytes in 0.000283 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 176 writes, 5764400 total bytes in 0.004503 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 73 writes, 2368208 total bytes in 0.001721 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 954920 total bytes in 0.000008 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 4356 writes, 14271236 total bytes in 0.114795 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 5 writes, 53184 total bytes in 0.000143 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 115104 writes, 115104 total bytes in 0.000008 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 1 writes, 502 total bytes in 0.000012 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 181 writes, 915950 total bytes in 0.000008 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 1 writes, 31880 total bytes in 0.000044 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 1 writes, 5780 total bytes in 0.000018 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 1633 writes, 1633 total bytes in 0.000008 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: fdfdo: 7918 reads, 24947488 total bytes in 6.132361 secs
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: Installing : oracle-instantclient-basic-21.0.0.0-1-eln.x86_64
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: D: closed db index
Sdp 23 06:49:31 aws055e56a2d01 /opt/awd/install/pr/privileged migration/package-helper: /var/lib/rpm/packages/

```

Server can't be access:



Database firewall still can use:

```

[oracle@standalone ~]$ !221
sqlplus avdfuser/12345@192.168.112.178:15221/pdb1

SQL*Plus: Release 19.0.0.0.0 - Production on Tue Sep 23 14:21:47 2025
Version 19.28.0.0.0

Copyright (c) 1982, 2025, Oracle. All rights reserved.

Last Successful login time: Mon Sep 22 2025 14:18:05 +07:00

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.28.0.0.0

SQL>
SQL> exit
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.28.0.0.0
[oracle@standalone ~]$ █

```

Finish upgrade on primary:

Reset Expired Password

supaudit, your password has expired.

old password

new password

confirm password

Submit

Your password has been reset successfully.

Oracle Audit Vault and Database Firewall

username

password

Login

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6. Verify That Audit Vault Agents and Host Monitor Agents Were Automatically Updated

On database target, 192.168.112.232

Agent:

```
[oracle@standalone bin]$ hostname -i
192.168.112.232
[oracle@standalone bin]$ ./agentctl --version

Audit Vault Agent : Release 20.15.0.0.0 (2025-09-23 09:08:01.010 +0000) - Production on Tue Sep 23 17:04:08 WIB 2025
Audit Vault Server Connection Details: (DESCRIPTION=(ENABLE=broken)(FAILOVER=on)(RETRY_COUNT=3)(CONNECT_TIMEOUT=60)(RETRY_DELAY=3)(TRANSPORT_CONNECT_TIMEOUT=3)(ADDRESS_LIST=(LOAD_BALANCE=on)(ADDRESS=(PROTOCOL=TCP)(HOST=192.168.112.121)(PORT=1521))(ADDRESS=(PROTOCOL=TCP)(HOST=192.168.112.120)(PORT=1521))))(CONNECT_DATA=(SERVICE_NAME=DBFWDB)))

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```

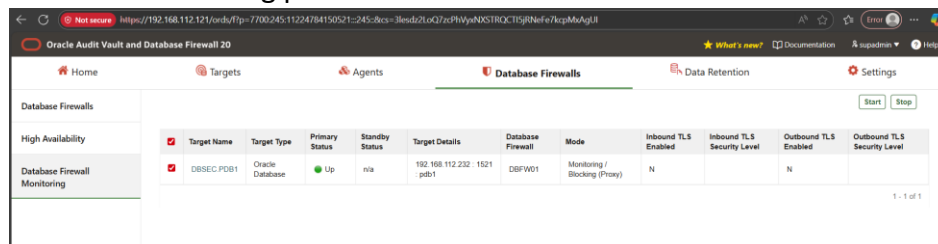
Host monitoring:

```
[oracle@standalone hm]$ cat version.txt
AV RELEASE VERSION = "20.15.0.0.0"
TIMESTAMP = "2025/08/18 12:18:20"
[oracle@standalone hm]$
[oracle@standalone hm]$ hostname -i
192.168.112.232
[oracle@standalone hm]$
```

7. Update Database Firewall (Primary Node)

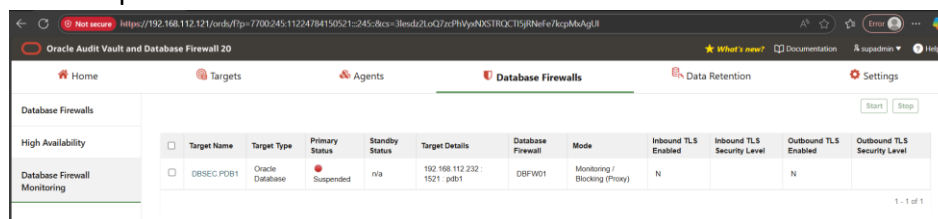
7.1. Stop all Database Firewall monitoring points.

- Log into the Audit Vault Server console as an administrator.
- Click the Database Firewalls tab.
- Click Database Firewall Monitoring in the left navigation menu.
- Select all monitoring points.



Target Name	Target Type	Primary Status	Standby Status	Target Details	Database Firewall	Mode	Inbound TLS Enabled	Inbound TLS Security Level	Outbound TLS Enabled	Outbound TLS Security Level
DBSEC-PDB1	Oracle Database	Up	n/a	192.168.112.232 : 1521 : pdb1	DBFW01	Monitoring / Blocking (Proxy)	N		N	

- Click Stop.



Target Name	Target Type	Primary Status	Standby Status	Target Details	Database Firewall	Mode	Inbound TLS Enabled	Inbound TLS Security Level	Outbound TLS Enabled	Outbound TLS Security Level
DBSEC-PDB1	Oracle Database	Suspended	n/a	192.168.112.232 : 1521 : pdb1	DBFW01	Monitoring / Blocking (Proxy)	N		N	

7.2.Run the Pre-upgrade RPM

```
[root@dbfw005056a08ff3 ~]# sha256sum avdf-pre-upgrade-20.15.0.0.0.zip
a4ac68c04648c96d86afd3fd0207e2fb577d993619b5c64703b8203aeb6e8393  avdf-pre-upgrade-20.15.0.0.0.zip
[root@dbfw005056a08ff3 ~]#
[root@dbfw005056a08ff3 ~]# unzip avdf-pre-upgrade-20.15.0.0.0.zip
Archive:  avdf-pre-upgrade-20.15.0.0.0.zip
  inflating: avdf-pre-upgrade-20.15.0.0.0-0_RELEASE.d.x86_64.rpm
[root@dbfw005056a08ff3 ~]#
[root@dbfw005056a08ff3 ~]# ll
total 340
-rw-----. 1 root   root    4786 Sep 12 13:09 anaconda-ks.cfg
-rw-r--r--. 1 root   root   170524 Sep 10 01:17 avdf-pre-upgrade-20.15.0.0.0-0_RELEASE.d.x86_64.rpm
-r--r--r--. 1 support support 158365 Sep 24 02:24 avdf-pre-upgrade-20.15.0.0.0.zip
-rw-----. 1 root   root    4973 Sep 12 13:13 original-ks.cfg
[root@dbfw005056a08ff3 ~]#
[root@dbfw005056a08ff3 ~]#
```

```
[root@dbfw005056a08ff3 ~]# rpm -i avdf-pre-upgrade-20.15.0.0.0-0_RELEASE.d.x86_64.rpm
warning: user oracle does not exist - using root
warning: group oinstall does not exist - using root
warning: group oinstall does not exist - using root
warning: group oinstall does not exist - using root
active
Checking upgrade preconditions
package avs is not installed
dbfw-20.12.0.0.0-0_240614.0100.x86_64
Verifying certificate date: /usr/local/dbfw/etc/ca.crt
Verifying certificate date: /etc/pki/tls/certs/localhost_internal.crt
Verifying certificate date: /usr/local/dbfw/etc/cert.crt
Checking upgrade preconditions
package avs is not installed
dbfw-20.12.0.0.0-0_240614.0100.x86_64
Verifying certificate date: /usr/local/dbfw/etc/ca.crt
Verifying certificate date: /etc/pki/tls/certs/localhost_internal.crt
Verifying certificate date: /usr/local/dbfw/etc/cert.crt
/usr/local/dbfw/bin/upgrade/oci-pki-check.rb: warning: argument of top-level return is ignored
Shutting down services.

AVDF::Installer::Upgrade::WarningPreconditions

Remove any schemas you have added to the database.

Additional schemas will cause the upgrade to fail.

SUCCESS:

The upgrade media can now be copied to '/var/dbfw/upgrade'.

The upgrade can then be started by running as root user:

    /usr/bin/avdf-upgrade

[root@dbfw005056a08ff3 ~]#
```

7.3.Transfer the ISO File to the Appliance

```
[root@standalone avdf]# scp avdf-pre-upgrade-20.15.0.0.0.iso support@192.168.112.178:/var/dbfw/software
The authenticity of host '192.168.112.178 (192.168.112.178)' can't be established.
ECDSA key fingerprint is SHA256:BNqjhJckPvePRGaSSMQd61ZX0KRBe0RHvctSTRSJhrM.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.112.178' (ECDSA) to the list of known hosts.
Database Firewall 20.12.0.0

DO NOT CHANGE ANY CONFIGURATIONS IN Database Firewall APPLIANCE WITHOUT GUIDANCE FROM
ORACLE SUPPORT. ANY CHANGES SHOULD BE TRACEABLE TO APPROPRIATE SR REFERENCE.
support@192.168.112.178's password:
avdf-pre-upgrade-20.15.0.0.0.iso
100% 16GB 63.5MB/s 04:21
[root@standalone avdf]#
```

```

/var/dbfw/software
[root@dbfw005056a08ff3 software]# ll
total 2064196
-rw-r--r--. 1 support support 2113732608 Sep 24 02:27 avdf-upgrade-20.15.0.0.0.iso
[root@dbfw005056a08ff3 software]# mv avdf-upgrade-20.15.0.0.0.iso /var/dbfw/upgrade
[root@dbfw005056a08ff3 software]# chown -R root:root /var/dbfw/upgrade/avdf-upgrade-20.15.0.0.0.iso
[root@dbfw005056a08ff3 software]#
[root@dbfw005056a08ff3 software]# ls -ltr /var/dbfw/upgrade/avdf-upgrade-20.15.0.0.0.iso
-rw-r--r--. 1 root root 17407113216 Sep 24 02:31 /var/dbfw/upgrade/avdf-upgrade-20.15.0.0.0.iso
[root@dbfw005056a08ff3 software]#

```

7.4.Start the Update Script

```

[root@dbfw005056a08ff3 ~]# /usr/bin/avdf-upgrade
'/Images' already mounted, Unmounting...
Unmounted /dev/mapper/vg_root-lv_images on /images
Please wait while validating SHA256 checksum for /var/dbfw/upgrade/avdf-upgrade-20.15.0.0.0.iso
Checksum validation successful for /var/dbfw/upgrade/avdf-upgrade-20.15.0.0.0.iso
Mounting /var/dbfw/upgrade/avdf-upgrade-20.15.0.0.0.iso on /images
mount: /images: WARNING: device write-protected, mounted read-only.
Successfully mounted /var/dbfw/upgrade/avdf-upgrade-20.15.0.0.0.iso on /images
/images/upgrade/lib/common.rb:1002: warning: mismatched indentations at 'rescue' with 'def' at 993
/images/upgrade/lib/chroot_upgrade.rb:337: warning: assigned but unused variable - ret

The following messages have important information about the upgrade process.

Power loss during upgrade may cause data loss. Do not power
off during upgrade.

Please review Note ID 2235931.1 for a current list of known issues.

The upgrade process is irreversible, please confirm 'y' to continue or 'n' to abort. [y/N]?
y
Verifying upgrade preconditions
package avs is not installed
dbfw-20.12.0.0.0-0_240614.0100.x86_64
Verifying certificate date: /usr/local/dbfw/etc/ca.crt
Verifying certificate date: /etc/pki/tls/certs/localhost_internal.crt
Verifying certificate date: /usr/local/dbfw/etc/cert.crt
mke2fs 1.45.6 (20-Mar-2020)
Discarding device blocks: done
Creating filesystem with 512000 4k blocks and 128000 inodes
Filesystem UUID: 6a38637b-384e-4407-9e17-faaeb40696eb
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

/images/upgrade/aux//oci-pki-check.rb: warning: argument of top-level return is ignored
1/11: Mounting filesystems (1)
2/11: Cleaning yum configuration
3/11: Cleaning old packages and files
Warning: failed to remove kernel-tools-libs
4/11: Upgrading kernel
5/11: Upgrading system
6/11: Cleaning platform packages repo
7/11: Adding required platform packages
8/11: Cleaning AVDF packages repo
9/11: Installing AVDF packages
10/11: Setting boot title
11/11: Setting final system status
Reboot now to continue the upgrade process.
Unmounted /var/dbfw/upgrade/avdf-upgrade-20.15.0.0.0.iso on /images
[root@dbfw005056a08ff3 ~]#

```

7.5.Restart the Applia

```

[root@dbfw005056a08ff3 ~]# reboot

Remote side unexpectedly closed network connection

Session stopped
- Press <Return> to exit tab
- Press R to restart session
- Press S to save terminal output to file

```

Checking background proses:

watch -n 5 "/opt/avdf/bin/privmigutl --status"

```
Every 5.0s: /opt/avdf/bin/privmigutl --status dbfw005056a08ff3: Wed Sep 24 03:00:31 2025
Migration set 'TOP' - succeeded
Last migration 'Switching to full runtime' - succeeded
No further action needed.
```

/opt/avdf/bin/privmigutl --history

```
[root@dbfw005056a08ff3 ~]# /opt/avdf/bin/privmigutl --history
2025-09-12 14:15:53 Migration TOP:1, package_helper["-c", "/media/avdf-install/dbfw-bootstrap/yum.conf", "--repo", "avdf-install-bootstrap", "-l", "boot
strap-os"] (as root) - succeeded
2025-09-12 14:15:53 Migration TOP:2, install-common-os-prcheck (as root) - succeeded
2025-09-12 14:15:53 Migration TOP:3, run-privileged-migrations["common-os"] (as root; retry permitted) - started
2025-09-12 14:15:54 Migration COMMON-OS:1, ntp (as root) - succeeded
2025-09-12 14:15:54 Migration COMMON-OS:2, selinux.rb (as root) - succeeded
2025-09-12 14:15:54 Migration COMMON-OS:3, stig-yum-config.rb (as root) - succeeded
2025-09-12 14:15:57 Migration COMMON-OS:4, configure-systemd-service.rb["avdf-services.yaml"] (as root) - succeeded
2025-09-12 14:15:57 Migration TOP:3, run-privileged-migrations["common-os"] (as root; retry permitted) - succeeded
2025-09-12 14:15:57 Migration TOP:4, install-common-os-postcheck (as root) - succeeded
2025-09-12 14:16:02 Migration TOP:5, package_helper["-c", "/media/avdf-install/dbfw-bootstrap/yum.conf", "--repo", "dbfw-install-bootstrap", "-l", "boot
strap-dbfw-dependencies"] (as root) - succeeded
2025-09-12 14:16:02 Migration TOP:6, install-dependencies-prcheck (as root) - succeeded
2025-09-12 14:16:02 Migration TOP:7, run-privileged-migrations["dependencies"] (as root; retry permitted) - started
2025-09-12 14:16:04 Migration DEPENDENCIES:1, update_conf_files.rb["--migration-target=dbfw.conf", "--mode=64", "--user=dbfw", "--group=dbfw"] (as root
) - succeeded
2025-09-12 14:16:03 Migration DEPENDENCIES:2, configure-syslog (as root) - succeeded
2025-09-12 14:16:04 Migration DEPENDENCIES:3, lvm-adjust-rl["--configlv-sanitisation"] (as root) - succeeded
2025-09-12 14:16:04 Migration DEPENDENCIES:4, import-network-settings (as root) - succeeded
2025-09-12 14:16:02 Migration DEPENDENCIES:5, preserve-ntp-config.rb (as root) - succeeded
2025-09-12 14:16:11 Migration DEPENDENCIES:6, configure-networking.rb["run-onlynmc-detection, iptables, disable_devices, create_nics, modify_nics, hosts, hos
tname, sshd, apache, certificate, routing, --no-restart-httpd"] (as root) - succeeded
2025-09-12 14:16:11 Migration DEPENDENCIES:7, oci-dependencies.rb (as root) - succeeded
2025-09-12 14:16:12 Migration DEPENDENCIES:8, yum["-y", "--disablerepo*", "-c", "/media/avdf-install/dbfw-ol-dependencies/yum.conf", "--enablerepo", "d
bfw-ol-dependencies", "clean", "all"] (as root) - succeeded
2025-09-12 14:16:13 Migration DEPENDENCIES:9, yum["-y", "--disablerepo*", "-c", "/media/avdf-install/dbfw-ol-dependencies/yum.conf", "--enablerepo", "d
bfw-ol-dependencies", "--groupinstall", "dbfw-ol-dependencies"] (as root) - succeeded
2025-09-12 14:16:13 Migration DEPENDENCIES:10, yum["-y", "--disablerepo*", "-c", "/media/avdf-install/dbfw-dependencies/yum.conf", "--enablerepo", "dbf
w-dependencies", "clean", "all"] (as root) - succeeded
2025-09-12 14:16:14 Migration DEPENDENCIES:11, yum["-y", "--disablerepo*", "-c", "/media/avdf-install/dbfw-dependencies/yum.conf", "--enablerepo", "dbf
w-dependencies", "upgrade", "dbfw"] (as root) - succeeded
2025-09-12 14:16:12 Migration DEPENDENCIES:12, yum["-y", "--disablerepo*", "-c", "/media/avdf-install/dbfw-dependencies/yum.conf", "--enablerepo", "dbf
w-dependencies", "clean", "all"] (as root) - succeeded
2025-09-12 14:16:12 Migration DEPENDENCIES:13, package_helper["-c", "/media/avdf-install/dbfw-dependencies/yum.conf", "--repo", "dbfw-dependencies", "-g
", "-l", "dbfw-dependencies,common-dependencies"] (as root) - succeeded
2025-09-12 14:16:13 Migration DEPENDENCIES:14, package_helper["-c", "/media/avdf-install/dbfw-dependencies/yum.conf", "--repo", "dbfw-dependencies", "-l
", "--oracle-unsentinel-basics"] (as root) - succeeded
2025-09-12 14:16:13 Migration TOP:7, run-privileged-migrations["dependencies"] (as root; retry permitted) - succeeded
2025-09-12 14:16:13 Migration TOP:8, install-dependencies-postcheck (as root) - succeeded
2025-09-12 14:16:13 Migration TOP:9, package_helper["-c", "/media/avdf-install/dbfw-bootstrap/yum.conf", "--repo", "dbfw-install-bootstrap", "-l", "boot
strap-dbfw-application"] (as root) - succeeded
2025-09-12 14:16:22 Migration TOP:10, install-application-prcheck (as root) - succeeded
2025-09-12 14:16:23 Migration TOP:11, run-privileged-migrations["application"] (as root; retry permitted) - started
2025-09-12 14:16:30 Migration APPLICATION:1, package_helper["-c", "/media/avdf-install/dbfw-application/yum.conf", "--repo", "dbfw-application", "--clea
n"] (as root) - succeeded
2025-09-12 14:16:22 Migration APPLICATION:2, package_helper["-c", "/media/avdf-install/dbfw-application/yum.conf", "--repo", "dbfw-application", "-g", "
", "--dbfw-application"] (as root) - succeeded
2025-09-12 14:16:22 Migration APPLICATION:3, run-privileged-migrations["common"] (as root; retry permitted) - started
2025-09-12 13:19:22 Migration COMMON:1, first (as nobody) - succeeded
2025-09-12 13:19:23 Migration COMMON:2, crypto-policy.rb (as root) - succeeded
2025-09-12 13:19:24 Migration COMMON:3, ssh-host-keys.rb (as root) - succeeded
2025-09-12 13:19:24 Migration COMMON:4, update-conf-files (as root) - succeeded
```

tail -f /var/log/messages

```
[root@dbfw005056a08ff3 ~]# tail -f /var/log/messages
Sep 24 02:59:55 dbfw005056a08ff3 systemd[1]: Started Session 4 of user support.
Sep 24 02:59:55 dbfw005056a08ff3 systemd-logind[1091]: New session 5 of user support.
Sep 24 02:59:55 dbfw005056a08ff3 systemd[1]: Started Session 5 of user support.
Sep 24 03:00:02 dbfw005056a08ff3 su: (to root) support on pts/1
Sep 24 03:00:46 dbfw005056a08ff3 systemd[1]: Starting AVDF Service - Update DBFW hostnames...
Sep 24 03:00:46 dbfw005056a08ff3 systemd[1]: Starting AVDF Service - DBFW log housekeeping...
Sep 24 03:00:47 dbfw005056a08ff3 systemd[1]: dbfw-housekeep-log.service: Succeeded.
Sep 24 03:00:47 dbfw005056a08ff3 systemd[1]: Started AVDF Service - DBFW log housekeeping.
Sep 24 03:00:47 dbfw005056a08ff3 systemd[1]: dbfw-hostname-lookup.service: Succeeded.
Sep 24 03:00:47 dbfw005056a08ff3 systemd[1]: Started AVDF Service - Update DBFW hostnames.
Sep 24 03:01:20 dbfw005056a08ff3 sshd[5557]: rexec line 19: Deprecated option UsePrivilegeSeparation
Sep 24 03:01:24 dbfw005056a08ff3 sshd[5567]: rexec line 19: Deprecated option UsePrivilegeSeparation
Sep 24 03:01:24 dbfw005056a08ff3 systemd-logind[1091]: New session 6 of user support.
Sep 24 03:01:24 dbfw005056a08ff3 systemd[1]: Started Session 6 of user support.
Sep 24 03:01:25 dbfw005056a08ff3 systemd-logind[1091]: New session 7 of user support.
Sep 24 03:01:25 dbfw005056a08ff3 systemd[1]: Started Session 7 of user support.
Sep 24 03:01:34 dbfw005056a08ff3 su: (to root) support on pts/2
Sep 24 03:01:52 dbfw005056a08ff3 systemd[1]: Starting AVDF Service - Update DBFW hostnames...
Sep 24 03:01:52 dbfw005056a08ff3 systemd[1]: Starting AVDF Service - DBFW log housekeeping...
Sep 24 03:01:53 dbfw005056a08ff3 systemd[1]: dbfw-housekeep-log.service: Succeeded.
Sep 24 03:01:53 dbfw005056a08ff3 systemd[1]: Started AVDF Service - DBFW log housekeeping.
Sep 24 03:01:53 dbfw005056a08ff3 systemd[1]: dbfw-hostname-lookup.service: Succeeded.
Sep 24 03:01:53 dbfw005056a08ff3 systemd[1]: Started AVDF Service - Update DBFW hostnames.
Sep 24 03:01:53 dbfw005056a08ff3 systemd[4502]: Starting Mark boot as successful...
Sep 24 03:01:53 dbfw005056a08ff3 systemd[4502]: Started Mark boot as successful.
Sep 24 03:02:56 dbfw005056a08ff3 systemd[1]: Starting AVDF Service - Update DBFW hostnames...
Sep 24 03:02:56 dbfw005056a08ff3 systemd[1]: Starting AVDF Service - DBFW log housekeeping...
Sep 24 03:02:57 dbfw005056a08ff3 systemd[1]: dbfw-housekeep-log.service: Succeeded.
Sep 24 03:02:57 dbfw005056a08ff3 systemd[1]: Started AVDF Service - DBFW log housekeeping.
Sep 24 03:02:57 dbfw005056a08ff3 systemd[1]: dbfw-hostname-lookup.service: Succeeded.
Sep 24 03:02:57 dbfw005056a08ff3 systemd[1]: Started AVDF Service - Update DBFW hostnames.
Sep 24 03:04:02 dbfw005056a08ff3 systemd[1]: Starting AVDF Service - Update DBFW hostnames...
Sep 24 03:04:02 dbfw005056a08ff3 systemd[1]: Starting AVDF Service - DBFW log housekeeping...
Sep 24 03:04:03 dbfw005056a08ff3 systemd[1]: dbfw-hostname-lookup.service: Succeeded.
Sep 24 03:04:03 dbfw005056a08ff3 systemd[1]: Started AVDF Service - Update DBFW hostnames.
Sep 24 03:04:03 dbfw005056a08ff3 systemd[1]: dbfw-housekeep-log.service: Succeeded.
Sep 24 03:04:03 dbfw005056a08ff3 systemd[1]: Started AVDF Service - DBFW log housekeeping.
Sep 24 03:05:12 dbfw005056a08ff3 systemd[1]: Starting AVDF Service - DBFW log housekeeping...
Sep 24 03:05:12 dbfw005056a08ff3 systemd[1]: Starting AVDF Service - Update DBFW hostnames...
Sep 24 03:05:13 dbfw005056a08ff3 systemd[1]: dbfw-housekeep-log.service: Succeeded.
Sep 24 03:05:13 dbfw005056a08ff3 systemd[1]: Started AVDF Service - DBFW log housekeeping.
Sep 24 03:05:13 dbfw005056a08ff3 systemd[1]: dbfw-hostname-lookup.service: Succeeded.
Sep 24 03:05:13 dbfw005056a08ff3 systemd[1]: Started AVDF Service - Update DBFW hostnames.
Sep 24 03:06:23 dbfw005056a08ff3 systemd[1]: Starting AVDF Service - Update DBFW hostnames...
```

tail -f /var/log/debug

```

[root@dbfw0505a08ff3 ~]# tail -f /var/log/debug
Sep 24 02:58:05 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - </struct>
Sep 24 02:58:05 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - </value>
Sep 24 02:58:05 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - </param>
Sep 24 02:58:05 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - </param>
Sep 24 02:58:05 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - Content-Type: text/xml
Sep 24 02:58:20 dbfw0505a08ff3 http-rack-server:4301: dbfw.admin_log_controller DEBUG - Method: GET, Path: /logs/1/list, Matched: download({instance="*", filename="1.txt"})
Sep 24 02:58:20 dbfw0505a08ff3 http-rack-server:4301: dbfw.admin_log_controller DEBUG - Method: GET, Path: /logs/1/list, Matched: download({instance="*", filename="1.txt"})
Sep 24 02:58:20 dbfw0505a08ff3 http-rack-server:4301: dbfw.admin_log_controller DEBUG - Method: GET, Path: /logs/1/list, Matched: download({instance="*", filename="1.txt"})
Sep 24 02:58:20 dbfw0505a08ff3 http-rack-server:4301: dbfw.admin_log_controller DEBUG - Method: GET, Path: /logs/1/list, Matched: download({instance="*", filename="1.txt"})
Sep 24 02:58:20 dbfw0505a08ff3 http-rack-server:4301: dbfw.admin_log_controller DEBUG - Method: GET, Path: /logs/1/list, Matched: download({instance="*", filename="1.txt"})
Sep 24 02:58:20 dbfw0505a08ff3 http-rack-server:4301: dbfw.admin_log_controller DEBUG - Method: GET, Path: /logs/1/list, Matched: download({instance="*", filename="1.txt"})
Sep 24 02:58:20 dbfw0505a08ff3 http-rack-server:4301: dbfw.admin_log_controller DEBUG - Method: GET, Path: /logs/1/list, Matched: download({instance="*", filename="1.txt"})
Sep 24 02:58:20 dbfw0505a08ff3 http-rack-server:4301: dbfw.admin_log_controller DEBUG - Method: GET, Path: /logs/1/list, Matched: download({instance="*", filename="1.txt"})
Sep 24 02:58:20 dbfw0505a08ff3 http-rack-server:4301: dbfw.admin_log_controller DEBUG - Method: GET, Path: /logs/1/list, Matched: download({instance="*", filename="1.txt"})
Sep 24 02:58:20 dbfw0505a08ff3 http-rack-server:4301: dbfw.admin_log_controller DEBUG - Method: GET, Path: /logs/1/list, Matched: download({instance="*", filename="1.txt"})
Sep 24 02:58:20 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - Method: GET, Path: /remote_access/fips, Matched: cll({cll="Fips"})
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - Request Body:
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - Request Arguments: 'GET DBFW:UI::Controller::FipsController show {cll="fips"}'
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - Service User:
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - CLI response:
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - [mode=auto, reboot_required=false]
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - Response Body:
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - "fail vera ions:1.8" encoding:"UTF-8" standalone:"yes"?>
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - <methodResponse>
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - <param>
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - <struct>
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - <member>
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - <name=enable/name>
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - <value>
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - <string=false/string>
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - </value>
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - </member>
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - <member>
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - <name=about_required/name>
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - <value>
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - <string=false/string>
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - </value>
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - </member>
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - </struct>
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - </param>
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - </methodResponse>
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - Content-Type:
Sep 24 02:58:08 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - Method: POST, Path: /remote_access/api, Matched: api({})
Sep 24 02:58:12 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - Request Body:
Sep 24 02:58:12 dbfw0505a08ff3 http-rack-server:4301: dbfw.http.remote_access DEBUG - <methodCall>

```

DBFW tidak bisa di akses dengan proxy port:

```

[oracle@standalone ~]$ !277
sqlplus a/vd/fuser/12345@192.168.112.178:15221/pdb1

SQL*Plus: Release 19.0.0.0.0 - Production on Wed Sep 24 09:46:35 2025
Version 19.28.0.0.0

Copyright (c) 1982, 2025, Oracle. All rights reserved.

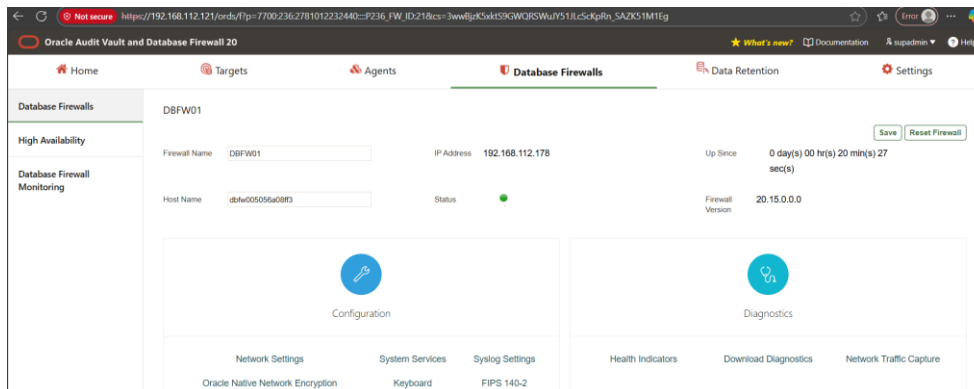
ERROR:
ORA-12541: TNS:no listener

Enter user-name:

```

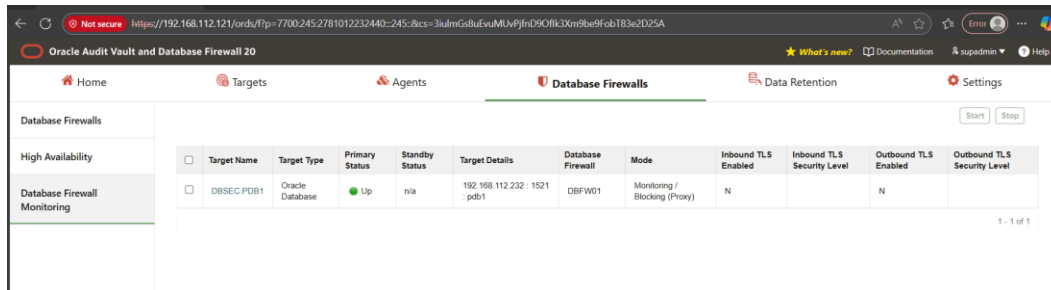
7.6.Re-register Database firewall

- Log in to the Audit Vault Server console as an administrator.
- Click the Database Firewalls tab.
- In the left navigation menu, Database Firewalls is selected by default and the page displays a list of configured Database Firewall instances.
- Select the Database Firewall instance that indicates a certificate error after the update.
- Click Reset Firewall.



7.7. Check proxy port Database firewall

Start monitoring point & status “up”



Target Name	Target Type	Primary Status	Standby Status	Target Details	Database Firewall	Mode	Inbound TLS Enabled	Inbound TLS Security Level	Outbound TLS Enabled	Outbound TLS Security Level
DBSEC.PDB1	Oracle Database	Up	na	192.168.112.232:1521 pdb1	DBFW01	Monitoring / Blocking (Proxy)	N		N	

Check connection dengan database firewall:

```
[oracle@standalones ~]$ sqlplus avdfuser/12345@192.168.112.178:15221/pdb1

SQL*Plus: Release 19.0.0.0.0 - Production on Wed Sep 24 10:22:44 2025
Version 19.28.0.0.0

Copyright (c) 1982, 2025, Oracle. All rights reserved.

Last Successful login time: Tue Sep 23 2025 14:21:47 +07:00

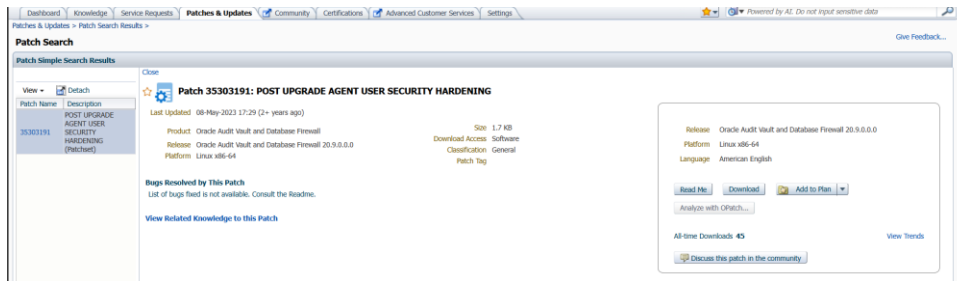
Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.28.0.0.0

SQL> exit
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.28.0.0.0
[oracle@standalones ~]$
```

8. Confirm the Update Process

8.1. Post Upgrade Agent User Security Hardening

Download patch 35303191



Follow the doc for unlock user avsys: “14.9 Unlocking and Locking the AVSYS User”

https://docs.oracle.com/en/database/oracle/audit-vault-database-firewall/20/sigad/user_accounts.html#GUID-648E1EB2-1324-4413-B77B-67BC4BDC8228

```
[root@avs005056a02dd1 ~]# su - oracle
Last login: Wed Sep 24 03:56:18 UTC 2025 on pts/0
The Oracle base has been set to /var/lib/oracle
[oracle@avs005056a02dd1 ~]$
[oracle@avs005056a02dd1 ~]$ cd revoke_user/
[oracle@avs005056a02dd1 revoke_user]$ ll
total 12
-rw-r--r--. 1 oracle oinstall 1722 Sep 24 03:55 p35303191_209000_Linux-x86-64.zip
-rw-r--r--. 1 oracle oinstall 2479 Apr 19 2023 README.txt
-rw-r--r--. 1 oracle oinstall 735 Apr 19 2023 revoke_privileges.sql
[oracle@avs005056a02dd1 revoke_user]$
```

```

[oracle@avs005056a02dd1 revoke_user]$ sqlplus avsys

SQL*Plus: Release 19.0.0.0.0 - Production on Wed Sep 24 03:57:11 2025
Version 19.28.0.0.0

Copyright (c) 1982, 2025, Oracle. All rights reserved.

Enter password:
Last Successful login time: Wed Sep 24 2025 03:55:12 +00:00

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.28.0.0.0

SQL> @revoke_privileges.sql
SQL> SET FEEDBACK 1
SQL> SET NUMWIDTH 10
SQL> SET LINESIZE 80
SQL> SET TRIMSPPOOL ON
SQL> SET TAB OFF
SQL> SET PAGESIZE 100
SQL>
SQL> BEGIN
2   execute immediate 'revoke execute on sys.dbms_lob from avbootstrap';
3   EXCEPTION
4   WHEN OTHERS THEN
5       IF SQLCODE = -1927 or SQLCODE = -01952 or SQLCODE = -01917 THEN
6           NULL;
7       ELSE
8           RAISE;
9       END IF;
10  END;
11  /

PL/SQL procedure successfully completed.

SQL>
SQL> BEGIN
2   execute immediate 'revoke select on AVSYS.FILE_REPOS from avbootstrap';
3   EXCEPTION
4   WHEN OTHERS THEN
5       IF SQLCODE = -1927 or SQLCODE = -01952 or SQLCODE = -01917 THEN
6           NULL;
7       ELSE
8           RAISE;
9       END IF;
10  END;
11  /

PL/SQL procedure successfully completed.

SQL> exit
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.28.0.0.0
[oracle@avs005056a02dd1 revoke_user]$

```

8.2.Clear Unused Kernels from Oracle Audit Vault and Database Firewall (optional)

Follow doc 2458154.1 for instructions to clear unused kernels

Dashboard
Knowledge
Service Requests
Patches & Updates
Community
Certifications
Advanced Customer Services
Settings

Give Feedback
You have been directed to this document based on an ID match. Alternatively, click here to search on this phrase. Hide

Post Upgrade Actions To Clear Unused Kernels From Oracle Audit Vault And Database Firewall (Doc ID 2458154.1)

To Bottom

In this Document

[Main Content](#)
[Purpose](#)
[Before you start](#)
[Steps to execute the script](#)

APPLIES TO:

Oracle Audit Vault and Database Firewall - Version 12.2.0.0 to 12.2.0.9 (Release 12.2 to 12.1)
Information in this document applies to any platform.

MAIN CONTENT

Purpose

This document contains the steps for clearing out unused kernels and related files.

The third-party scanning tools report unused kernel packages installed that are not used. The script [kernel_cleanup.sh](#) runs on the system and provides suggestions about the files and packages to be removed.

The script performs the following:

- Runs brief sanity tests to see if it is appropriate to continue
- Looks for files not owned by any packages in /boot
- Looks for unused kernels
- Advises the user which commands to run to remove anything extra
- Advises the user on how to verify that the system has the correct set of packages

Note: This script does not update any kernels. It only removes unused ones. To receive updates, apply the latest bundle patch.

Was this document helpful?

☐ Yes
☐ No

Document Details

Type: PRODUCT NEWS
Status: ARCHIVED
Last Major Update: 16-Oct-2018
Last Update: 09-Jun-2023
Language: English

Related Products

Oracle Audit Vault and Database Firewall

Information Centers

Index of Oracle Database Information Centers [1568943.2]
インフォメーション・センター・データベース [1946305.2]
Enterprise Manager 日本語ドキュメント [1946305.2]

Document References

No References available for this document.

Recently Viewed

Post Upgrade Actions To Clear Unused Kernels From Oracle Audit Vault And Database Firewall

```
[root@avs005056a02dd1 ~]# df -h
Filesystem                                Size  Used Avail Use% Mounted on
devtmpfs                                  5.9G   0  5.9G   0% /dev
tmpfs                                     5.9G  41M  5.8G   1% /run
tmpfs                                     5.9G   0  5.9G   0% /sys/fs/cgroup
/dev/mapper/vg_root-lv_ol8root            6.5G  4.3G  1.9G  70% /
/dev/mapper/vg_root-lv_home              943M  192K  878M   1% /home
/dev/mapper/vg_root-lv_var_tmp            5.5G  196K  5.2G   1% /var/tmp
/dev/mapper/vg_root-lv_local_dbfw         943M  277M  661M  32% /usr/local/dbfw
/dev/mapper/vg_root-lv_var_log            5.5G  382M  4.9G   6% /var/log
/dev/mapper/vg_root-lv_var_opt_oracle_ahf  15G  298M  14G   3% /var/opt/oracle/ahf
/dev/mapper/vg_root-lv_tmp                1.9G  552K  1.8G   1% /tmp
/dev/mapper/vg_root-lv_var_dbfw           2.8G   64K  2.6G   1% /var/dbfw
/dev/mapper/vg_root-lv_opt               943M  199M  680M  23% /opt
/dev/mapper/vg_root-lv_oracle             58G   29G   26G  53% /var/lib/oracle
tmpfs                                     1.0G   0  1.0G   0% /usr/local/dbfw/volatile
/dev/mapper/vg_root-lv_local_dbfw_tmp     6.5G  104K  6.1G   1% /usr/local/dbfw/tmp
/dev/mapper/vg_root-lv_opt_oracle_ahf     4.9G   1.4G  3.2G  31% /opt/oracle/ahf
/dev/mapper/vg_root-lv_images             463M   16K  434M   1% /images
/dev/mapper/vg_root-lv_var_log_audit      1.9G  331M  1.5G  19% /var/log/audit
/dev/mapper/vg_root-lv_var_dbfw_upgrade   29G   17G   11G  60% /var/dbfw/upgrade
tmpfs                                     12G   1.1G   11G  10% /dev/shm
/dev/sda2                                974M  238M  677M  26% /boot
/dev/sda1                                200M  6.0M  194M   3% /boot/efi
tmpfs                                     1.2G   0  1.2G   0% /run/user/507
tmpfs                                     1.2G   0  1.2G   0% /run/user/503
tmpfs                                     1.2G   0  1.2G   0% /run/user/1000
[root@avs005056a02dd1 ~]#
[root@avs005056a02dd1 ~]# mount /boot
mount: /boot: /dev/sda2 already mounted on /boot.
[root@avs005056a02dd1 ~]#
[root@avs005056a02dd1 ~]# ruby ./kernel-cleanup.rb
# FAILED
undefined method 'untaint' for an instance of String
[root@avs005056a02dd1 ~]#
```

8.3. Check the Observer Status After Updating to Oracle AVDF 20.7 or Later for High Availability

```
[root@avs005056a02dd1 ~]# su - oracle
Last login: Wed Sep 24 04:03:42 UTC 2025 on pts/0
The Oracle base has been set to /var/lib/oracle
[oracle@avs005056a02dd1 ~]$
[oracle@avs005056a02dd1 ~]$
[oracle@avs005056a02dd1 ~]$ dgmgrl /
DGMGRL for Linux: Release 19.0.0.0 - Production on Wed Sep 24 04:09:46 2025
Version 19.28.0.0.0

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Welcome to DGMGRL, type "help" for information.
Connected to "DBFWDB_HA1"
Connected as SYSDB.
DGMGRL>
DGMGRL> show observer

Configuration - DBFWDB

Fast-Start Failover:      ENABLED

Primary:                  DBFWDB_HA1
Active Target:            DBFWDB_HA2

Observer "avs005056a02204" - Master

Host Name:                avs005056a02204
Last Ping to Primary:     2 seconds ago
Last Ping to Target:      2 seconds ago

Observer "avs005056a02dd1" - Backup

Host Name:                avs005056a02dd1
Last Ping to Primary:     0 seconds ago
Last Ping to Target:      2 seconds ago

DGMGRL>
DGMGRL> exit
[oracle@avs005056a02dd1 ~]$
```


8.4.Start audit trails

Home

Targets

Agents

Database Firewalls

Data Retention

Settings

Targets

Database Discovery

Audit Trails

Target Groups

Access Rights

Start Stop Delete Downtime Move

Go Actions

	Target	Trail Location	Trail Type	Status	Target Type	Agent	Last Started	Data Collected Until	Trail Autostart Details	Error Message
☐	DBSEC.PDB1	UNIFIED_AUDIT_TRAIL	TABLE	Stopped	Oracle Database	DBSEC STANDALONES	9/24/2025 11:19:44 AM	9/23/2025 1:13:16 PM	Status: Enabled Attempts: 0 Last start: 9/24/2025 11:19:44 AM	

1 - 1

Start:

Home

Targets

Agents

Database Firewalls

Data Retention

Settings

Targets

Database Discovery

Audit Trails

Target Groups

Access Rights

Start Stop Delete Downtime Move

Go Actions

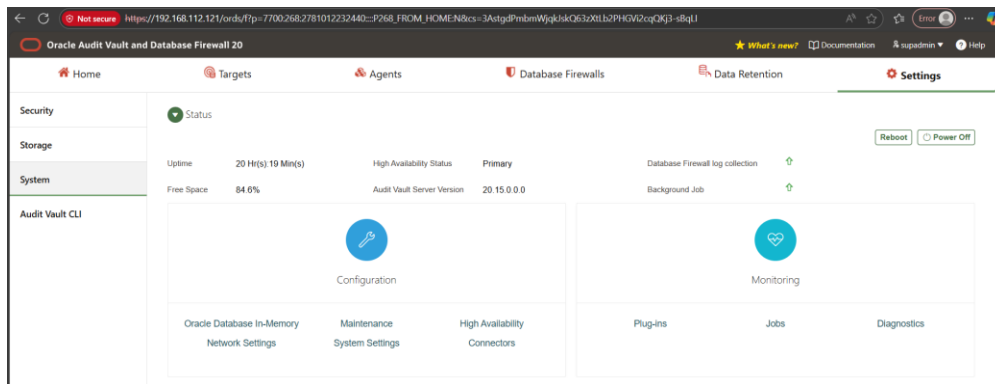
	Target	Trail Location	Trail Type	Status	Target Type	Agent	Last Started	Data Collected Until	Trail Autostart Details	Error Message
☐	DBSEC.PDB1	UNIFIED_AUDIT_TRAIL	TABLE	Idle	Oracle Database	DBSEC STANDALONES	9/24/2025 11:19:44 AM	9/24/2025 11:27:07 AM	Status: Enabled Attempts: 0 Last start: 9/24/2025 11:19:44 AM	

1 - 1

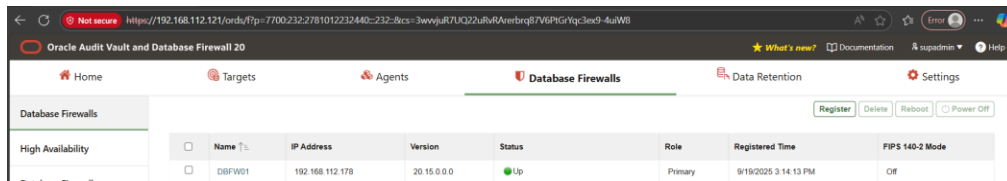
9. Precheck After upgrade

After:

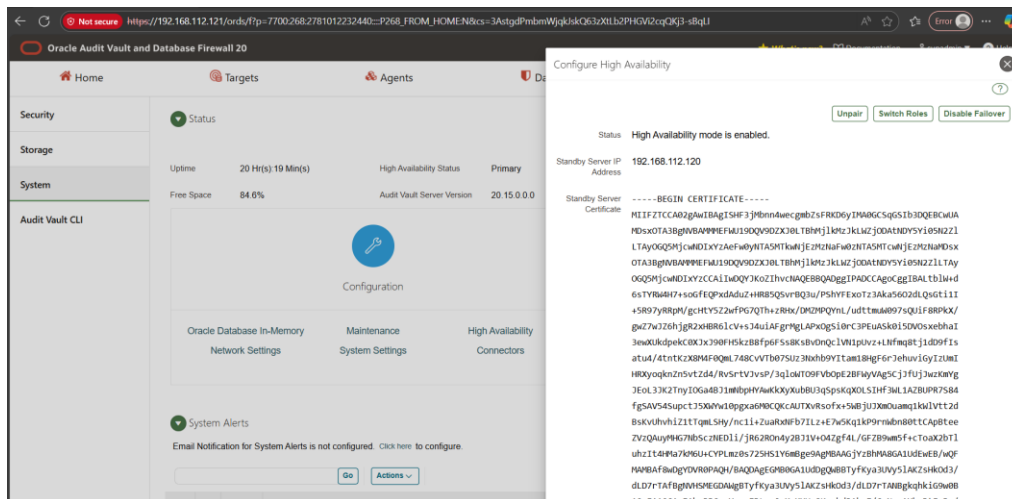
AVS Version:



DBFW Version:



HA AVS:



```

[oracle@avs005056a02dd1 ~]$ !11
/usr/local/dbfw/bin/setup_ha.rb --status
HA mode: PRIMARY
HA server 1: 192.168.112.121
HA server 2: 192.168.112.120
Unique database name: DBFWDB HA1
Current database role: PRIMARY
Data guard broker: ENABLED
Data guard observer: YES
Current log mode: ARCHIVELOG
Logging forced: YES
Flashback mode: YES
Current open mode: READ WRITE
Switchover status: TO STANDBY
Automatic failover: ENABLED
Failover status: TARGET UNDER LAG LIMIT
Missing listener services: NONE
Archive destination: ENABLE
Recovery mode: MANAGED REAL TIME APPLY
FRA size: 11,811,160,064
FRA used: 9,210,691,584
FRA reclaimable: 8,100,249,600
FRA available: 10,700,718,080
Gap status: NO GAP
Archived sequence: 64
Applied sequence: 63
Apply lag: 0:00:14
[oracle@avs005056a02dd1 ~]$
[oracle@avs005056a02dd1 ~]$ date
Wed Sep 24 04:24:37 UTC 2025
[oracle@avs005056a02dd1 ~]$

```

```

[oracle@avs005056a02204 ~]$ !34
/usr/local/dbfw/bin/setup_ha.rb --status
HA mode: STANDBY
HA server 1: 192.168.112.121
HA server 2: 192.168.112.120
Unique database name: DBFWDB HA2
Current database role: PHYSICAL STANDBY
Data guard broker: ENABLED
Data guard observer: YES
Current log mode: ARCHIVELOG
Logging forced: YES
Flashback mode: YES
Current open mode: MOUNTED
Switchover status: NOT ALLOWED
Automatic failover: ENABLED
Failover status: TARGET UNDER LAG LIMIT
Missing listener services: NONE
Archive destination: RESET
Recovery mode: MANAGED REAL TIME APPLY
FRA size: 11,811,160,064
FRA used: 8,755,609,600
FRA reclaimable: 7,531,921,408
FRA available: 10,587,471,872
[oracle@avs005056a02204 ~]$
[oracle@avs005056a02204 ~]$ date
Wed Sep 24 04:21:42 UTC 2025
[oracle@avs005056a02204 ~]$

```

DB Target:

Agent Version:

```

[oracle@standalones bin]$
[oracle@standalones bin]$ hostname -i
192.168.112.232
[oracle@standalones bin]$
[oracle@standalones bin]$ ./agentctl --version

Audit Vault Agent : Release 20.15.0.0.0 (2025-09-23 09:08:01.010 +0000) - Production on Tue Sep 23 17:04:08 WIB 2025
Audit Vault Server Connection Details: (DESCRIPTION=(ENABLE=BROKEN))(FAILOVER=on)(RETRY_COUNT=3)(CONNECT_TIMEOUT=60)(RETRY_DELAY=3)(TRANSPORT_CONNECT_TIMEOUT=3)(ADDRESS_LIST=
((LOAD_BALANCE=on)(ADDRESS=(PROTOCOL=TCP)(HOST=192.168.112.121)(PORT=1521)))(ADDRESS=(PROTOCOL=TCP)(HOST=192.168.112.120)(PORT=1521)))(CONNECT_DATA=(SERVICE_NAME=DBFWDB
.DBFWDB)))

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```

Host monitoring Version:

```

[oracle@standalones hm]$
[oracle@standalones hm]$ cat version.txt
AV_RELEASE_VERSION = "20.15.0.0.0"
TIMESTAMP = "2025/08/18 12:18:20"
[oracle@standalones hm]$
[oracle@standalones hm]$ hostname -i
192.168.112.232
[oracle@standalones hm]$

```

Database firewall diagnostics:

Oracle Audit Vault and Database Firewall 20

Home Targets

Database Firewalls DBFW01

High Availability

Database Firewall Monitoring

Firewall Name DBFW01

Host Name dbfw005056a08b3

Network Settings

Oracle Native Network

Health Indicators

- Certificates
- Configuration
- Database Firewall Monitoring
- Filesystem
- Networking
- System

Diagnostics

Diagnostics	Result
controller_connection	●
exists_var_dbfw_va_1_etc_appliance_conf	●
monitor_is_running	●
monitoring_processes	●
permissions_on_policy_file_oracle_policy_36_json	●
permissions_on_policy_file_oracle_policy_7_json	●
stand_is_running	●
unexpected_policy_file	●
unexpected_va_directory_entries	●
writable_by_user_dbfw_var_dbfw_va_1_log	●

1 - 10 of 10

Data Retention

Up Since 0 day(s) 00 hr(s) 27 min(s) 14 sec(s)

Firewall Version 20.15.0.0.0

Diagnostics

Download Diagnostics Network Traffic Capture

Save Reset Firewall

Oracle Audit Vault and Database Firewall 20

Home Targets

Database Firewalls DBFW01

High Availability

Database Firewall Monitoring

Firewall Name DBFW01

Host Name dbfw005056a08b3

Network Settings

Oracle Native Network

Health Indicators

- Certificates
- Configuration
- Database Firewall Monitoring
- Filesystem
- Networking
- System

Diagnostics

Diagnostics	Result
dris	●
dbfw_conf_net_service_map	●
gateway	●
netfilter_rules	●
netfilter_address	●
network_mask	●
state_of_device_ens192	●
that_ipv6_is_disabled	●

1 - 8 of 8

Data Retention

Up Since 0 day(s) 00 hr(s) 27 min(s) 14 sec(s)

Firewall Version 20.15.0.0.0

Diagnostics

Download Diagnostics Network Traffic Capture

Save Reset Firewall

Dashboard:

Oracle Audit Vault and Database Firewall 20

Home Targets Agents Database Firewalls Data Retention Settings

System Alerts

10

Category Count

High Availability	2
Storage	2
Password	2

1 - 3 Next >

Click to configure email notification

Targets

1

Type Count

Oracle Database	1
-----------------	---

1 - 1

Audit Collection

Agents

1

Platform Count

Linux x86_64	1
--------------	---

1 - 1

Audit Trails

1

Type Count

Tails	1
-------	---

1 - 1

Database Firewall Monitoring

Database Firewalls

2

Up Down

Monitoring Points

1

Up Down

Host Monitor

0

Mode Count

Monitoring (Backup Proxy)	1
---------------------------	---

Collection Summary

45,000

10,000

0

18-Sep 19-Sep 20-Sep 21-Sep 22-Sep 23-Sep 24-Sep

Total Records Collected 89k

Average Collection Rate in last 1 week 12 records/min

Targets with High Collection Rate in last 1 week 1

Jobs Summary

User jobs in last 1 week Failed: 10, Success: 11

System jobs in last 1 week Failed: 1, Success: 82

Data Retention Summary

Automatic Records Archiving Disabled Click here to Enable

Total size of Event data collected < 1 GB

Total size of Event data in Archive 0 GB

Data Retention Job Status Next Run Time 9/25/2020 11:48:38 AM

Retention Date 0

Datafiles available for moving to a remote location 0

System Overview

Audit Vault Server Version	20.15.0.0.0
High Availability Status	Primary
Uptime	21 hrs 28 mins
CPU Utilization	19% (2 CPUs)
Disk Space Utilization	Filesystem: 18.4% used (Total 251.7 GB, Used 39.73 GB) Database Storage: 82.7% used (Total 48 GB, Used 28 GB)
Memory Utilization	60.2% used (Total 12 GB)
Network Traffic Log Collection	Up
PODFULS Report Engine	Up
Alerts Engine	Up

Appendix:

Testing register DBFW 20.12 integrate with AVS 20.15: show alert need upgrade to the latest

Register Database Firewall

1 error has occurred.

• OAV-47704: Database Firewall DBFW03 is not on the latest version. Upgrade to the latest.

Copy Command

Name

DBFW03

IP Address

192.168.112.180

Cancel

Register